

Direct observation of the neural computations underlying a single decision

Reviewed Preprint

v2 • July 29, 2024

Revised by authors

Reviewed Preprint

v1 • October 12, 2023

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Abstract

Neurobiological investigations of perceptual decision-making have furnished the first glimpse of a flexible cognitive process at the level of single neurons (Shadlen and Newsome, 1996; Shadlen and Kiani, 2013). Neurons in the parietal and prefrontal cortex (Kim and Shadlen, 1999; Romo et al., 2004; Hernández et al., 2002; Ding and Gold, 2012) are thought to represent the accumulation of noisy evidence, acquired over time, leading to a decision. Neural recordings averaged over many decisions have provided support for the deterministic rise in activity to a termination bound (Roitman and Shadlen, 2002). Critically, it is the unobserved stochastic component that is thought to confer variability in both choice and decision time (Gold and Shadlen, 2007). Here, we elucidate this drift-diffusion signal on individual decisions. We recorded simultaneously from hundreds of neurons in the lateral intraparietal cortex (LIP) of monkeys while they made decisions about the direction of random dot motion. We show that a single scalar quantity, derived from the weighted sum of the population activity, represents a combination of deterministic drift and stochastic diffusion. Moreover, we provide direct support for the hypothesis that this drift-diffusion signal approximates the quantity responsible for the variability in choice and reaction times. The population-derived signals rely on a small subset of neurons with response fields that overlap the choice targets. These neurons represent the integral of noisy evidence. Another subset of direction-selective neurons with response fields that overlap the motion stimulus appear to represent the integrand. This parsimonious architecture would escape detection by state-space analyses, absent a clear hypothesis.

eLife assessment

This **fundamental** work quantifies the stochastic dynamics of neural population activity in the lateral intraparietal area (LIP) of the macaque monkey brain during single perceptual decisions. These single-trial dynamics have been subject to intense debate in neuroscience, and they have significant implications for modelling decision-making in various fields including neuroscience and psychology. Through a combination of state-of-the-art recordings from many LIP neurons and theory-driven data analyses, the authors provide **convincing** evidence for the notion that single-trial neural population dynamics in LIP encode the decision variable postulated by the drift-diffusion model of decision-making.

<https://doi.org/10.7554/eLife.90859.2.sa3>

Introduction

Neural signals in the mammalian cortex are notoriously noisy. They manifest as a sequence of action potentials (spikes) that approximate non-stationary Poisson point processes. Therefore, to characterize the signal produced by a neuron, electrophysiologists typically combine the spike times from many repetitions, or trials, relative to the time of an event (e.g., stimulus onset), to yield the average firing rate of the neuron as a function of time. Such trial-averaged firing rates are the main staple of systems neuroscience. They are the source of knowledge about spatial selectivity (e.g., receptive fields), feature selectivity (e.g., direction of motion, faces vs. other objects), and even cognitive signals associated with working memory, anticipation, attention, motor planning, and decision making. But there is an important limitation.

Trial averages suppress signals that vary independently across trials (here and throughout, trial average and across-trial average refer to the mean of signal values, over all specified trials at the same time, t , relative to a trial event [e.g., motion onset]). In many cognitive tasks, such as difficult decisions, the variable component of the signal is the most interesting, because it is this component that is thought to explain the variable choice and response time. This variability is thought to arise from a decision process that accumulates noisy evidence in favor of the alternatives and terminates when the accumulated evidence for one alternative, termed the decision variable (DV), reaches a terminating bound. The DV is stochastic because the integral of noisy samples of evidence is biased Brownian motion (or drift-diffusion) and this leads to a stochastic choice and response time on each decision. However, the stochastic part of this signal is suppressed by averaging across trials. We will use the term drift-diffusion because it is the expression most commonly applied in models of decision making (Ratcliff and Rouder, 1998 [↗](#); Gold and Shadlen, 2007 [↗](#)), and we will consider the noise part—that is, diffusion—as the signal of interest.

In the setting of difficult perceptual decisions, studied here, bounded drift-diffusion reconciles the relationship between decision speed and accuracy. It also explains the trial-averaged firing rates of neurons in the lateral intraparietal area (LIP) that represent the action used by monkeys to indicate their choice. These firing rate averages show motion-dependent, ramping activity that reflects the direction and strength of motion, consistent with the drift component of drift-diffusion (Roitman and Shadlen, 2002 [↗](#)). Up to now, however, the diffusion component has not been observed, owing to averaging.

There is thus a missing link between the mathematical characterization of the decision process and its realization in neural circuits, leaving open the possibility that drift-diffusion dynamics do not underlie LIP activity (e.g., Latimer et al., 2015 [DOI](#)), or emerge only at the level of the population, without explicit representation by single neurons. We reasoned that these and other alternatives to drift-diffusion could be adjudicated if it were possible to resolve the DV giving rise to a single decision.

This stratagem is now feasible, owing to the development of high-density Neuropixels probes, which are capable of recording from deep sulci in the primate brain. Here we provide the first direct evidence for a drift-diffusion process underlying single decisions. We recorded simultaneously from up to 203 neurons in the lateral intraparietal area (LIP) while monkeys made perceptual decisions about the direction of dynamic random dot motion (Newsome et al., 1989 [DOI](#); Gold and Shadlen, 2007 [DOI](#)). Using a variety of dimensionality reduction techniques, we show that a drift-diffusion signal can be detected in such populations on individual trials. Moreover, this signal satisfies the criteria for a DV that controls the choice and reaction time. Notably, the signal of interest is dominated by a small subpopulation of neurons with response fields that overlap one of the choice targets, consistent with earlier single neuron studies (e.g., Shadlen and Newsome, 1996 [DOI](#); Roitman and Shadlen, 2002 [DOI](#); Churchland et al., 2011 [DOI](#); Gold and Shadlen, 2007 [DOI](#)).

Results

Two monkeys made perceptual decisions, reported by an eye movement, about the net direction of dynamic random dot motion (**Fig. 1a** [DOI](#)). We measured the speed and accuracy of these decisions as a function of motion strength (**Fig. 1b** [DOI](#), circles). The choice probabilities and the distribution of reaction times (RT) are well described (**Fig. 1b** [DOI](#), traces) by a bounded drift-diffusion model (**Fig. 1c** [DOI](#)). On 50% of the trials, a brief (100 ms) pulse of weak leftward or rightward motion was presented at a random time. The influence of these pulses on choice and RT further supports the assertion that the choices and RTs arose through a process of integration of noisy samples of evidence to a stopping bound (**Fig. S1** [DOI](#); Stine et al., 2020 [DOI](#), 2023 [DOI](#); Hyafil et al., 2023 [DOI](#)). In addition to the main task, the monkeys performed two control tasks: instructed, delayed saccades to peripheral targets and passive viewing of random dot motion (see Methods). These control tasks served to identify, *post hoc*, neurons with response fields that overlap the choice target in the hemifield contralateral to the recording site (T_{in}^{con}), neurons with response fields that overlap the other choice target (T_{in}^{ips}), and neurons with response fields that overlap the random-dot motion stimulus (M_{in} ; **Table 2** [DOI](#)).

We recorded simultaneously from populations of neurons in area LIP, using newly developed macaque Neuropixels probes while monkeys performed these tasks. The data set comprises eight sessions from two monkeys (1696–2894 trials per session; **Table 2** [DOI](#)). Our primary goal was to identify activity in LIP that relates to the decision variable, a theoretical quantity that determines the choice and reaction time on each trial. To achieve this, we formed weighted averages from all neurons in the sample population, including those with response fields that do not overlap a choice target or the motion stimulus. We used several strategies to assign this vector of weights, which we refer to as a *coding direction* in the neuronal state space. The projection of the spiking activity from the population of neurons onto the vector of weights gives rise to a scalar function of time, $S^x(t)$, where the superscript x labels the strategy. We focus on such one-dimensional projections because of the long-standing hypothesis that the DV is drift-diffusion, which is a scalar function of time.

We first developed a targeted strategy that would reproduce the well-known coherence-dependent ramping activity evident in the across-trial averages. This strategy applies regression to best approximate a linear ramp, on each trial, i , that terminates with a saccade to the choice-target

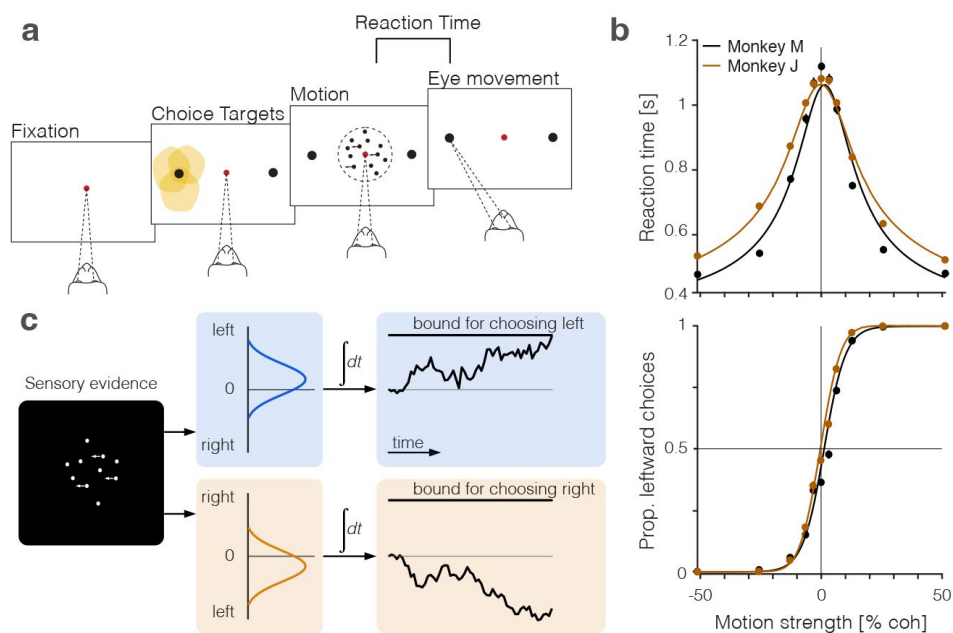


Figure 1.

Perceptual decisions are explained by the accumulation of noisy evidence to a stopping bound.

a, Random dot motion discrimination task. The monkey fixates a central point. After a delay, two peripheral targets appear, followed by the random dot motion. When ready, the monkey reports the net direction of motion by making an eye movement to the corresponding target. Yellow shading indicates the response fields of a subset of neurons in area LIP that we refer to as T_{in} neurons (*target in response field*). **b**, Mean reaction times (top) and proportion of leftward choices (bottom) plotted as a function of motion strength and direction, indicated by the sign of the coherence: positive is leftward. Data (circles) are from all sessions from monkey M (black, 9684 trials) and monkey J (brown, 8142 trials). Solid lines are fits of a bounded drift-diffusion model. **c**, Drift-diffusion model. The decision process is depicted as a race between two accumulators: one integrating momentary evidence for left; the other for right. The momentary samples of evidence are sequential samples from a pair negatively correlated Normal distributions with opposite means ($\rho = -0.71$). The decision is terminated when one accumulator reaches its positive bound. The example depicts leftward motion leading to a leftward decision.

contralateral to the hemisphere of the LIP recordings. The ramps are defined on the epoch spanning the decision time: from $t_0 = 0.2$ s after motion onset to $t_1 = 0.05$ s before saccade initiation (black lines in **Fig. S2**). The epoch is motivated by many previous studies (e.g., Gold and Shadlen, 2007, and see below). Each ramp begins at $f_i(t_0) = -1$ and ends at $f_i(t_1) = 1$. The ramp approximates the expectation—conditional on the choice and response time—of the deterministic components of the drift-diffusion signal which can incorporate (i) the deterministic drift, (ii) a time-dependent but evidence-independent urgency signal (Drugowitsch et al., 2012), and (iii) a dynamic bias signal (Hanks et al., 2011). It can also be viewed as an approximation to firing rates averaged across trials and grouped by contraversive choice and RT quantile (e.g., Roitman and Shadlen, 2002, see also **Fig. S3**). Importantly, the fit is not guided by an assumption of an underlying diffusion process. That is, the ramp coding direction is agnostic to the underlying processes whose averages approximate ramps. The weights derived from these regression fits specify a ramp-coding direction in the state space defined by the population of neurons in the session. The single-trial signal, $S^{\text{ramp}}(t)$, is rendered as the projection of the population firing rates onto this coding direction.

The left side of **Fig. 2a** shows single-trial activity rendered by this strategy. The right side of the figure shows the averages of the single-trial responses grouped by signed coherence and aligned to motion onset or response time (saccade initiation). These averaged traces exhibit features of the firing rate averages in previous studies of single neurons in LIP (e.g., see Roitman and Shadlen, 2002; Gold and Shadlen, 2007, and **Fig. S3**). They begin to diverge as a function of the direction and strength of motion approximately 200 ms after the onset of motion. The traces converge near the time of saccadic response to the contralateral choice-target such that the coherence dependence is absent or greatly diminished. Coherence dependence remains evident through the initiation of saccades to the right (ipsilateral) target, consistent with a race architecture—between negatively correlated accumulators—depicted in **Fig. 1c**.

We complemented this regression strategy with Principal Component Analysis (PCA) and use the first PC (PC1), which explains $44 \pm 7\%$ of the variance (mean $\pm$ s.e. across sessions) of the activity between 200 and 600 ms from motion onset (see Methods). This coding direction renders single trial signals, $S^{\text{PC1}}(t)$ (**Fig. 2b**). In a third strategy, we consider the mean activity of neurons with response fields that overlapped the contralateral choice target ($T_{\text{in}}^{\text{con}}$ neurons), which were the focus of previous single-neuron studies (e.g., Shadlen and Newsome, 1996; Platt and Glimcher, 1999; Roitman and Shadlen, 2002). In those studies, the task was modified so that one of the choice targets was placed in the neural response field, whereas here we identify neurons *post hoc* with response fields that happen to overlap the contralateral choice target. This difference probably accounts for the lower firing rates of the $T_{\text{in}}^{\text{con}}$ neurons studied here. **Fig. 2c** shows single-trial and across-trial averages from these $T_{\text{in}}^{\text{con}}$ neurons. They too render signals, $S_{\text{Tin}}^{\text{con}}(t)$, similar to those derived from the full population. The $T_{\text{in}}^{\text{con}}$ neurons thus furnish a third coding direction defined by a vector of identical positive weights assigned to all $T_{\text{in}}^{\text{con}}$ neurons and 0 for all other neurons in the population. The emboldened single-trial traces in **Fig. 2** left correspond to the same trials rendered by the three coding directions. It is not difficult to tell which are the corresponding traces, an observation that speaks to their similarity, and the same is true for the averages. We will expand on this observation in what follows.

The averages show the deterministic *drift* component of the hypothesized drift-diffusion process, with the slope varying monotonically with the signed motion strength (**Fig. 2** right). The rise begins to saturate as a consequence of the putative termination bound—a combination of dropout of trials that are about to terminate and the effect on the distribution of possible diffusion paths imposed by the very existence of a stopping bound. This saturation is evident earlier on trials with stronger motion, hence shorter RT, on average. The positive buildup rate on the 0% coherence motion represents the time-dependent, evidence-independent signal that is thought to reflect the cost of time. It leads to termination even if the evidence is weak, equivalent to collapsing stopping

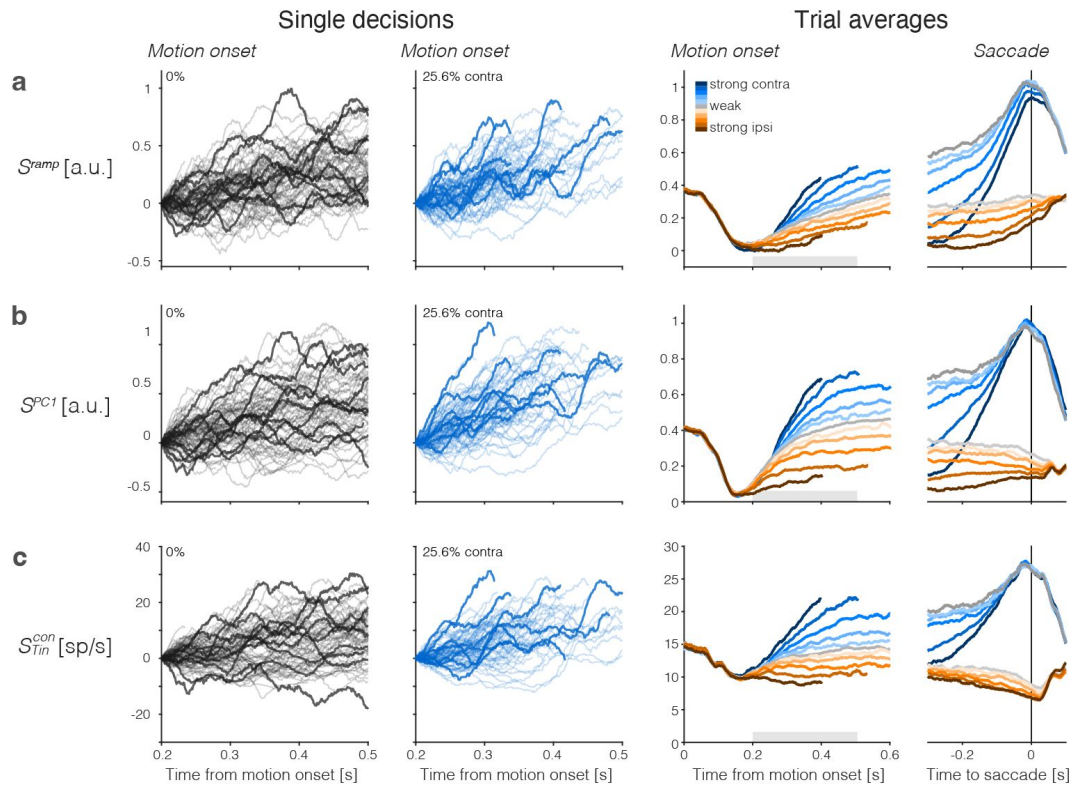


Figure 2.

Population responses from LIP approximate drift-diffusion.

Rows show three types of population signals. The left columns show representative single-trial firing rates during the first 300 ms of evidence accumulation using two motion strengths: 0% and 25.6% coherence toward the left (contralateral) choice target. For visualization, single-trial traces were baseline corrected by subtracting the activity in a 50 ms window around 200 ms. We highlight several trials with thick traces (same trials in a–c). The right columns show the across-trial average responses for each coherence and direction. Motion strength and direction are indicated by color (legend) and aligned to motion onset (left) or saccade initiation (right). The gray bars under the motion-aligned averages indicate the 300 ms epoch used in the display of the single-trial responses (left panels). The epoch begins when LIP first registers a signal related to the strength and direction of motion. Except for saccade-aligned response, trials are cut off 100 ms before saccade initiation. Error trials are excluded from the saccade-aligned averages, only. **a**, Ramp coding direction. The weight vector is established by regression to ramps from -1 to $+1$ over the period of putative integration, from 200 ms after motion onset to 100 ms before saccade initiation (see Fig. S2). Only trials ending in left (contralateral) choices are used in the regression. **b**, First principal component (PC1) coding direction. **c**, Average firing rates of the subset of neurons that represent the left (contralateral) target. The weight vector consists of $\frac{1}{N}$ for each of the N T_{in}^{con} neurons and 0 for all other neurons. Note the similarity of both the single-trial traces and the response averages produced by the different weighting strategies.

bounds in traditional, symmetric drift-diffusion models (Drugowitsch et al., 2012). Removal of this *urgency* signal, $u(t)$, from the non-zero coherence traces renders the positive and negative coherence averages symmetric relative to zero on the ordinate (Fig. S4).

The single-trial responses in Fig. 2 do not look like the averages but instead approximate drift-diffusion. We focus on the epoch from 200 to 500 (or 600) ms from motion onset—that is, the first 300 (or 400) ms of the period in which the averages reflect the integration of evidence. Some traces are cut off before the end of the epoch because a saccade occurred 100 ms later on the trial. However, most 0% coherence trials continue beyond 500 ms (median RT > 600 ms). The single-trial traces do not rise monotonically as a function of time but meander and tend to spread apart from each other vertically. For unbounded diffusion, the variance would increase linearly, but as just mentioned, the existence of an upper stopping bound and the limited range of firing rates (e.g., non-negative) renders the function sublinear at later times (Fig. 3a). The autocorrelation between an early and a later sample from the same diffusion trace is also clearly specified for unbounded diffusion. The theoretical values shown in Fig. 3b & c are the autocorrelations of unbounded diffusion processes that are smoothed identically to the neural signals (see Methods and Appendix). The autocorrelations in the data follow a strikingly similar pattern. These observations support the assertion that the coherence-dependent (ramp-like) firing rate averages observed in previous studies of area LIP are composed of stochastic drift-diffusion processes on single trials.

Single-trial drift-diffusion signals approximate the decision variable

We next evaluate the hypothesis that the drift-diffusion signal, $S^x(t)$, is the decision variable that controls the choice and response time. We have identified several coding directions that produce candidate DVs, and as we will see below, there are also other coding directions of interest that can be derived from the population. Additionally, PCA indicates that the dimensionality of the data is low, but greater than one (participation ratio = 4.4 ± 1.3 ; Mazzucato et al., 2016; Gao et al., 2017). Therefore, one might wonder whether it is sensible to assume that the DV can be approximated by a scalar measure arising from a single coding direction as opposed to a higher dimensional representation. Two decoding exercises are adduced to support the assumption.

We constructed a logistic decoder of choice using each neuron's spike counts in 50 ms bins between 100 and 500 ms after motion onset. As shown in Fig. 4a, this *What*-decoder (orange) predicts choice as accurately as a decoder of simulated data from a drift-diffusion model (black) using parameters derived from fits to the monkeys' choice and RT data (see Methods). The simulation establishes a rough estimate of the decoding accuracy that can be achieved, given the stochastic nature of the choice, were we granted access to the drift-diffusion signal that actually determines the decision. In this analysis, the decoder can use a different vector of weights at each point in time (*time-dependent* coding directions; see Peixoto et al., 2021). However, if the representation of the decision variable in LIP is one-dimensional, then a decoder trained at one time should perform well when tested at a different time. The red curve in Fig. 4a shows the performance of a *What*-decoder with a *fixed training-time* (450 ms after motion onset; red arrow). This decoder performs nearly as well as the decoder trained at each time bin. The heatmap (Fig. 4b) generalizes this observation. It shows two main features for all times $300 < t < 500$ ms (dashed box). First, unsurprisingly, for a *What* choice decoder trained on data at one time $t = x$, the predictions improve as the testing time advances (the decoding accuracy increases along any vertical) as more evidence is accrued. Second, and more importantly, decoders tested at time $t = y$ perform similarly, independent of when they were trained (there is little variation in decoding accuracy along any horizontal). This observation suggests that a single vector of weights may suffice to decode the choice from the population response.

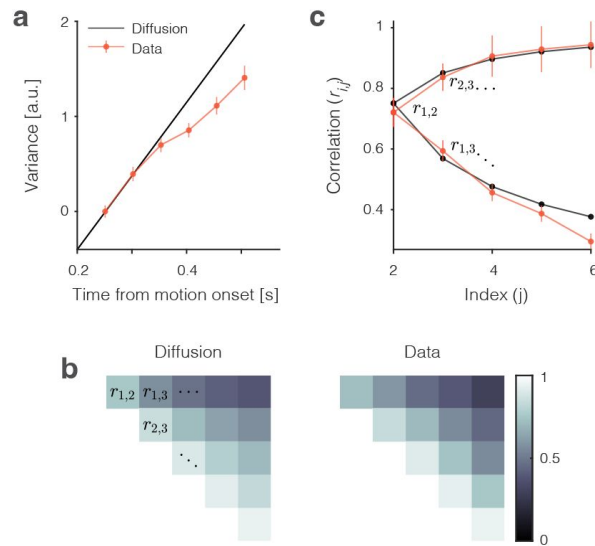


Figure 3.

Variance and autocorrelation of the single trial signals.

The analyses here are based on samples of S^{ramp} at six time points during the first 300 ms of putative integration, using all 0% and $\pm 3.2\%$ coherence trials ($N = 5927$). Samples are separated by the width of the boxcar filter (51 ms), beginning at $t_1 = 226$ ms. **a**, Variance increases as a function of time. The measure of variance is normalized so that it is 1 for the first sample. Error bars are s.e. (bootstrap). **b**, Autocorrelation of samples as a function of time and lag approximate the values expected from diffusion. The upper triangular portion of the 6×6 correlation matrix for unbounded diffusion (r_{ij}) is represented by brightness). The values from the data (S^{ramp}) are similar (right). **c**, Nine of the 15 autocorrelation terms in **b** permit a more direct comparison of theory and data. The lower limb of the C-shaped function shows the decay in r_{ij} as a function of lag ($j - i$). This is the top row of **b**. The upper limb shows the increase in r_{ij} as a function of time (for fixed lag). This is the lower diagonal in **b**. Error bars are s.e. (bootstrap). Note that the autocorrelations incorporate a free parameter, $\phi \leq 1$, that serves to correct for an unknown fraction of the measured variance that is not explained by diffusion (see Methods).

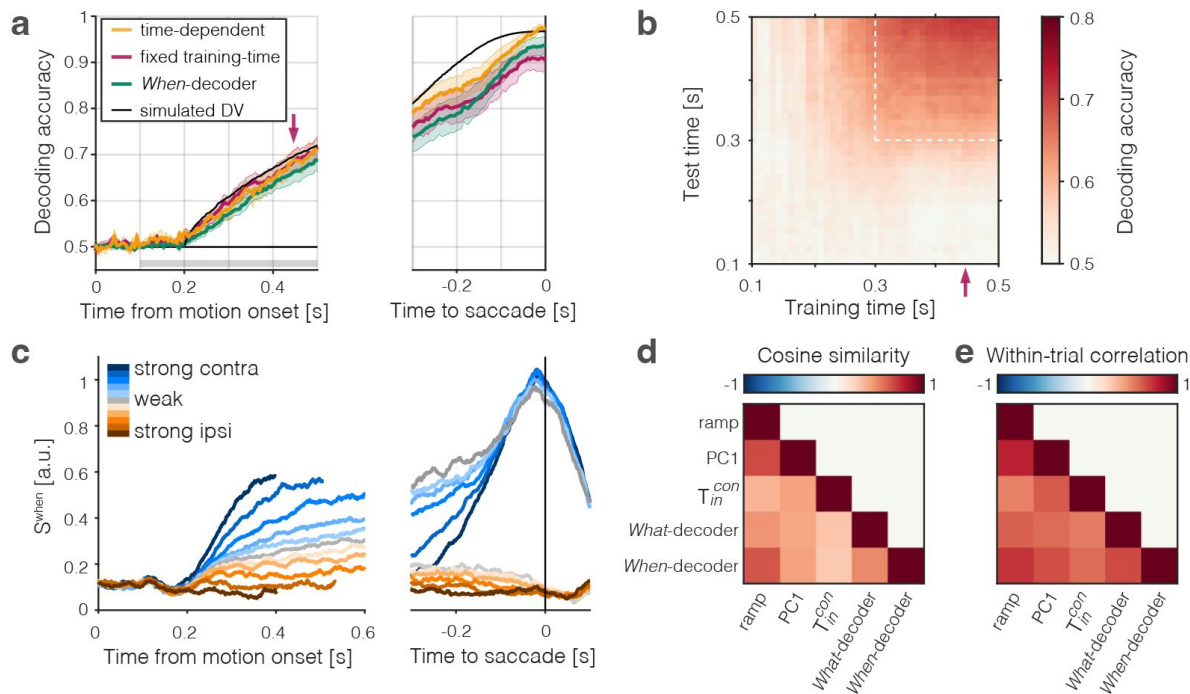


Figure 4.

The population signal predictive of choice and RT is approximately one-dimensional.

Two binary decoders were trained to predict the choice (*What*-decoder) and its time (*When*-decoder) using the population responses in each session. The *When*-decoder predicts whether a saccadic response to the contralateral target will occur in the next 150 ms, but critically, its accuracy is evaluated based on its ability to predict choice. **a**, Choice decoding accuracy plotted as a function of time from motion onset (left) and time to saccadic choice (right). Values are averages across sessions. The *What*-decoder is either trained at the time point at which it is evaluated (time-dependent decoder, orange) or at the single time point indicated by the red arrow ($t = 450$ ms after motion onset; *fixed training-time* decoder, red). Both training procedures achieve high levels of accuracy. The *When*-decoder is trained to capture the time of response only on trials terminating with a left (contraversive) choice. The coding direction identified by this approach nonetheless predicts choice (green) nearly as well as the *fixed training-time* *What*-decoder. The black trace shows the accuracy of a *What*-decoder trained on simulated signals using a drift-diffusion model that approximates the behavioral data in **Fig. 1**. Error bars signify s.e.m. across sessions. The gray bar shows the epoch depicted in the next panel. **b**, The heatmap shows the accuracy of a decoder trained at times along the abscissa and tested at times along the ordinate. Time is relative to motion onset (gray shading in **a**). In addition to training at $t = 450$ ms, the decoder can be trained at any time from $300 < t < 500$ ms (dashed box) and achieve the same level of accuracy when tested at any single test time. The orange and red traces in **a** correspond to the main diagonal ($x = y$) and the column marked by the red arrow, respectively. **c**, Trial-averaged activity rendered by the projection of the population responses along the *When* coding direction, S^{When} . Same conventions as in **Fig. 2**. **d**, Cosine similarity of five coding directions. The heatmap shows the mean values across the sessions, arranged like the lower triangular portion of a correlation matrix. Cosine similarities are significantly greater than zero for all comparisons (all $p < 0.001$, t-test). **e**, Correlation of single-trial diffusion traces. The Pearson correlations are calculated from ordered pairs of $\{S_i^x(t), S_i^y(t)\}$, where $S_i(t)$ are the detrended signals rendered by coding directions, x and y , on trial i . The detrending removes trial-averaged means for each signed coherence, leaving only the diffusion component. Reported correlations are significantly greater than zero for all pairs of coding directions and sessions (all $p < 10^{-23}$, t-test, see Methods). The variability in cosine similarity and within-trial correlation across sessions is portrayed in **Fig. S8**.

The second decoder is trained to predict whether a saccade to the contralateral choice target will be initiated in the next 150 ms. This *When*-decoder is trained by logistic regression to predict a binary output: 1 at all time-points that are within 150 ms of an upcoming saccade and 0 elsewhere (Fig. S6). We validated the *When*-decoder by computing the area under an ROC (AUC) using the held-out (odd) trials (mean AUC over all time points: 0.84), but this is tangential to our goal. Although the *When*-decoder was trained only to predict the time of saccades, our rationale for developing this decoder was to test whether the *When* coding direction can be used to predict the choice. The green trace in Fig. 4a shows the accuracy of $S^{\text{When}}(t)$ to predict the choice. The performance is almost identical to the choice decoder, despite being trained on a temporal feature of trials ending in the same left choice. This feat is explained by the similarity of signals produced by the *When*- and other coding directions. Note the similarity of the trial averaged S^{When} signals displayed in Fig. 4c to those in Fig. 2 (see also Fig. S7a, right). Indeed, the cosine similarity between the *When* and *Ramp* coding directions is 0.67 ± 0.03 (Fig. 4d). In light of this, it is not surprising that the weighting vectors derived from both the *What*- and *When*-decoders also render single-trial drift-diffusion traces that resemble each other and those rendered by other coding directions (Fig. 4e). Together these analyses support the assertion that the DV is likely to be captured by a single dimension, consistent with Ganguli et al. (2008).

If the one-dimensional signals, $S^X(t)$, approximate the decision variable, they should explain the variability of choice and reaction time for trials sharing the same direction and motion strength. Specifically, (i) early samples of $S^X(t)$ should be predictive of choice and correlate inversely with the RT on trials that result in contraversive (leftward) choices, (ii) later samples ought to predict choice better and correlate more strongly (negatively) with RT than earlier samples, and (iii) later samples should contain the information present in the earlier samples and thus mediate (i.e., reduce the leverage) of the earlier samples on choice and RT. Each of these predictions is borne out by the data.

The analyses depicted in Fig. 5 allow us to visualize the influence of the single trial signals, $S^X(t)$, on the choice and RT on that trial. We focus on the early epoch of evidence accumulation (200–550 ms after random dot motion onset) and restrict the analyses to decisions with RT ≥ 670 ms and coherence $\leq 6.4\%$. The RT restriction eliminates 17% of the eligible trials. Larger values of $S^X(t)$ are associated with a larger probability of a left (contraversive) choice and a shorter RT, hence negative correlation between $S^X(t)$ and RT. We use the term, leverage, to describe the strength of both of these associations. The leverage on choice (Fig. 5a, black traces) is the contribution of $S^X(t)$ to the log odds of a left choice, after accounting for the motion strength and direction (i.e., the coefficients, $\beta_1(t)$ in Eq. 8). The leverage on RT (Fig. 5b) is the Pearson correlation between $S^X(t)$ and the RT on that trial, after accounting for the effect of motion strength and direction on S^X and RT (See Methods). The leverage is evident from the earliest sign of evidence accumulation, 200 ms after motion onset, and its magnitude increases as a function of time, as evidence accrues (Fig. 5, top). The filled circle to the right of the traces in each graph shows the leverage of S^X at $t = 550$ ms, which is 120 ms before any of the included trials have terminated. Both observations are consistent with the hypothesis that S^X represents the integral of noisy evidence used to form and terminate the decision. Two control analyses demonstrate that the degree of leverage on choice and RT do not arise by chance: (i) random coding directions in state space produce negligible leverage (Fig. S10, top), and (ii) breaking the trial-by-trial correspondence between neural activity and behavior eliminates all leverage (see Reviewer Figure 1 in reply to peer review).

Importantly, the leverage at earlier times is mediated by the later sample at $t = 550$ ms. The blue traces in all graphs show the remaining leverage, once this later sample is allowed to explain the choice and RT—by including, respectively, an additional term in the logistic regression (Eq. 9) and calculating the partial correlation, conditional on $S^X(t = 0.55)$. We assessed statistical significance of the mediation statistics, ξ^{Ch} and ξ^{RT} (Eqs. 7 and 10) in each session for the three signals shown in Fig. 5 using a bootstrap procedure (see Methods, below Eq. 10). Mediation is significant in 47 of the 48 comparisons (all $p < 0.023$, median $p < 10^{-317}$). The one non-

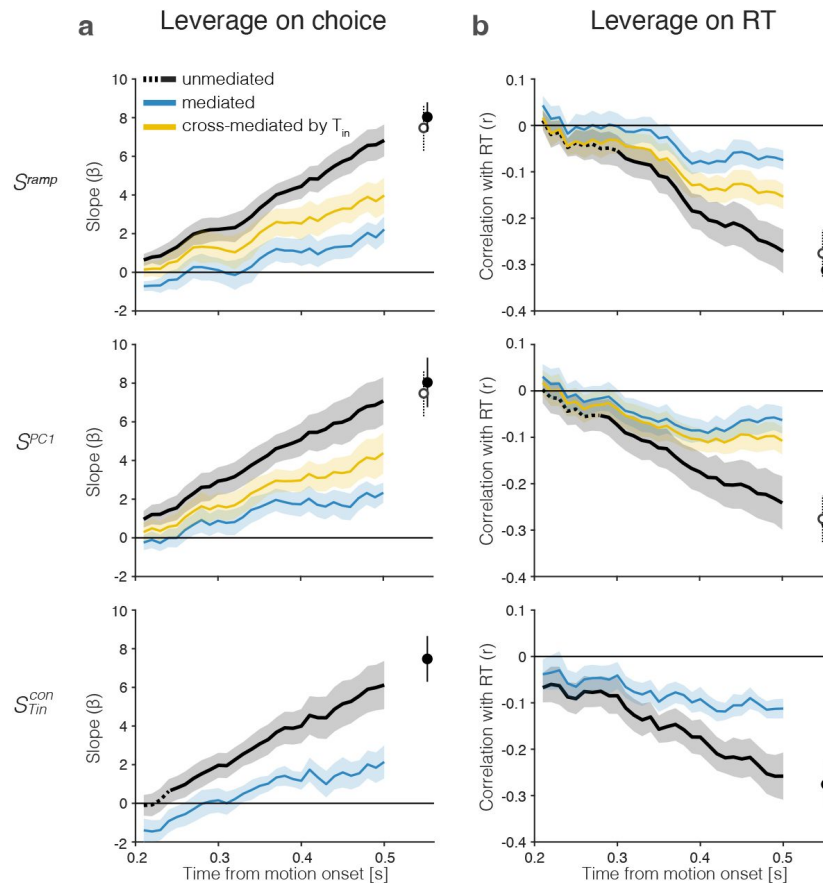


Figure 5.

The drift-diffusion signal approximates the decision variable.

The graphs show the leverage of single-trial drift-diffusion signals on choice and RT using only trials with $RT \geq 0.67$ s. Rows correspond to the same coding directions as in [Fig. 2](#). The graphs also demonstrate a reduction of the leverage of the samples at $t \leq 0.5$ s by a later sample of the signal at $t = 0.55$ s. Error bars are s.e.m. across sessions. **a**, Leverage of single-trial drift-diffusion signals on choice. Leverage is the value of β_1 , the coefficient that multiplies $S^x(t)$ in [Eq. 8](#). The black traces show the increase in leverage as a function of time. The dashed linestyle at the left end of three of the traces indicate values that are not statistically significant ($p > 0.05$, bootstrap shuffle test, see Methods). Filled symbols show the leverage at $t = 0.55$ s. The blue curve (*mediated*) shows the leverage when the later sample is included in the regression ([Eq. 9](#)). Open symbols show the leverage of S_{Tin}^{con} at $t = 0.55$ s (same value as the filled symbol in *bottom row*). The yellow curves (top and middle rows) show the leverage of the *cross-mediated* signal by $S_{Tin}^{con}(0.55)$. For all three signals, $S^x(t)$, leverage at $t = 0.4$ s is significantly mediated by $S^x(0.55)$ and cross-mediated by $S_{Tin}^{con}(0.55)$ (all $p < 10^{-5}$, Paired Samples t-test). **b**, Leverage of single-trial drift-diffusion signals on response time. Same conventions as in **a**. Leverage is the correlation between $S^x(t)$ and RT. The mediated leverage is the partial correlation, given the later sample. For all three signals, $S^x(t)$, leverage at $t = 0.4$ s is significantly mediated by $S^x(0.55)$ and cross-mediated by $S_{Tin}^{con}(0.55)$ (all $p < 10^{-8}$, Paired Samples t-test).

significant comparison is ξ^{Ch} for $S_{\text{Tin}}^{\text{con}}$ in session 2 ($p = 0.73$). The mediation is significant when this comparison is included in the combined data ($p < 10^{-9}$, Paired Samples t-test). The stark decrease in leverage is consistent with one-dimensional diffusion in which later values of the signal contain the information in the earlier samples plus what has accrued in the interim. Had we recorded from all the neurons that represent the DV, we would expect the mediation to be complete (e.g., partial correlation = 0). However, our recorded population is only a fraction of the entire population. Indeed, the observed degree of mediation is similar to values obtained from simulations of weakly correlated, noisy neurons (**Fig. S10**, *bottom*).

There is one additional noteworthy observation in **Fig. 5** that highlights the importance of the $T_{\text{in}}^{\text{con}}$ neurons. The top and middle rows (S^{ramp} and S^{PC1}) contain a second, open symbol, which is simply a copy of the filled symbol from the bottom row ($S_{\text{Tin}}^{\text{con}}$). The yellow traces show significant cross-mediation of S^{ramp} and S^{PC1} by the sample, $S_{\text{Tin}}^{\text{con}}(t = 0.55)$ (all $p < 0.05$; median $p < 10^{-268}$; bootstrap as above). This signal, carried by 9–21% of the neurons, mediates signals produced by the full population of 54–203 neurons nearly as strongly as S^{ramp} and S^{PC1} mediate themselves. The observation suggests that minimal leverage is gained by sophisticated analyses of the full neuronal state space compared to a simple average of $T_{\text{in}}^{\text{con}}$ neurons. This is both reassuring and disquieting: reassuring because the $T_{\text{in}}^{\text{con}}$ neurons compose the dominant projection from LIP to the portions of the superior colliculus and the frontal eye field involved in the generation of saccades toward the contralateral choice target (Paré and Wurtz, 1998; Ferraina et al., 2002); disquieting because the functional relevance of these neurons is not revealed by the other coding directions. The weights assigned to the $T_{\text{in}}^{\text{con}}$ neurons span all percentiles (mean IQR: 49–96; mean 71st percentile, $AUC = 0.74 \pm 0.05$) in the ramp coding direction. They contribute disproportionately to PC1 and the *What*- and *When*-decoders but not enough to stand out based on their weights. Indeed, the ability to predict that a neuron is $T_{\text{in}}^{\text{con}}$ from its weight or percentile is remarkably poor (**Fig. S9**).

These observations support the idea that the single-trial signals, S^{ramp} , S^{PC1} and $S_{\text{Tin}}^{\text{con}}$, approximate the DV used by the monkey to make its decision. In **Fig. S11** we show that the S^{What} and S^{When} coding directions achieve qualitatively similar results. Moreover, a late sample from $S^{\text{X}}(t)$ mediates the earlier correlation with RT and choice of signals rendered by other coding directions, $S^{\text{Y}}(t)$, at earlier times (**Fig. S11**). Such cross-mediation is consistent with the high degree of cosine similarity between the coding directions (**Fig. 4d**). The observation suggests that the decision variable is a prominent signal in LIP, discoverable by a variety of strategies, and consistent with the idea that it is one-dimensional. In **Fig. S12**, we show that linear and nonlinear decoders achieve similar performance, which argues against a non-linear embedding of the decision variable in the population activity.

Activity of direction-selective neurons in area LIP resembles momentary evidence

Up to now, we have focused our analyses on resolving the DV on single trials, paying little attention to how it is computed or to other signals that may be present in the LIP population. The drift-diffusion signal approximates the accumulation, or integral, of the noisy momentary evidence—a signal approximating the difference in the firing rates of direction-selective (DS) neurons with opposing direction preferences (e.g. in area MT; Britten et al., 1996). DS neurons, with properties similar to neurons in MT, have also been identified in area LIP (Freedman and Assad, 2006; Shushruth et al., 2018; Fanini and Assad, 2009; Bollimunta and Ditterich, 2012), where they are proposed to play a role in motion categorization (Freedman and Assad, 2011). We hypothesize that such neurons might participate in routing information from DS neurons in MT/MST to those in LIP that contain a choice-target in their response fields.

We identified such DS neurons using a passive motion viewing task (Fig. 6a–b, left). Neurons preferring leftward or rightward motion constitute 5–10% of the neurons in our sample populations Table 2. Fig. 6 shows the average firing rates of 51 leftward-preferring neurons (M_{in}^{left} , Fig. 6a) and 26 rightward-preferring neurons (M_{in}^{right} , Fig. 6b) under passive motion viewing and decision-making. The separation of the two traces in the passive viewing task is guaranteed because we used this task to identify the DS neurons. It is notable, however, that the DS is first evident about 100 ms after the onset of random dot motion, and this latency is also apparent in mean firing rates grouped by signed coherence during decision making (Fig. 6a,b right). The activity of DS neurons is modulated by both the direction and strength of motion. However, unlike the T_{in} neurons, the traces associated with different motion strengths are mostly parallel to one another and do not reach a common level of activity before the saccadic eye movement (*i.e.*, they do not signal decision termination).

In addition to their shorter onset latency, the direction-selectivity of M_{in} neurons precedes the choice-selectivity of T_{in}^{con} neurons by ~ 100 ms (Fig. 6c). The responses bear similarity to DS neurons in area MT. Such neurons are known to exhibit choice-dependent activity insofar as they furnish the noisy evidence that is integrated to form the decision (Britten et al., 1996; Shadlen et al., 1996). We computed putative single trial direction signals by averaging the responses from the left- and right-preferring DS neurons, respectively. The resulting signals, $S_{Min}^{left}(t)$ and $S_{Min}^{right}(t)$, have weak leverage on choice, but the leverage does not increase as a function of time (Fig. 6d, left). This is what would be expected if the M_{in} neurons represent the noisy momentary evidence as opposed to the accumulation thereof (Mazurek et al., 2003). We failed to detect a correlation between RT and either S_{Min} signal (Fig. 6d, right). This is surprising, but it could be explained by lack of power—a combination of small numbers of M_{in} neurons, narrow sample windows (50 ms boxcar) and the focus on the long RT trials. Indeed, we found a weak but statistically significant negative correlation between RT and the difference in leftward vs. rightward signals, averaged over the epoch $0.1 \leq t \leq 0.4$ s from motion onset ($p = 0.0004$; $\mathcal{H}_0: \rho \geq 0$, see Methods).

We considered the hypothesis that these DS signals are integrated by the T_{in}^{con} neurons to form the DV. The heatmap in Fig. 6e supports this hypothesis. On each trial, we formed the ordered pairs, $\{x, y\}$, where $x = \tilde{S}_{Min}^{left}(t_x) - \tilde{S}_{Min}^{right}(t_x)$ and $y = \tilde{S}_{T_{in}}^{con}(t_y)$. The tilde in these expressions indicates the use of standardized residual values, for each signed motion strength. The heatmap shows the correlation of these quantities across trials. If the hypothesis were true, the correlations should be positive for $t_y > t_x$ when $t_x > 100$ ms and $t_y > 200$ ms, and if the operation approximates integration, the level of correlation should be consistent at all lags, $t_y - t_x > 100$ ms. The correlations are significant in the epoch of interest, and they differ significantly from the average correlations in the rest of the graph (*i.e.*, $t_x < 100$, $t_y < 200$, or $t_y < t_x$; $p < 0.0001$, permutation test). Although correlative, the observation is consistent with the idea that evidence integration occurs within area LIP, rather than inherited from another brain area (Zhang et al., 2022; Bollimunta and Ditterich, 2012).

Discussion

We have observed a neural representation of the stochastic process that gives rise to a single decision. This is the elusive drift-diffusion signal that has long been thought to determine the variable choice and response time in the perceptual task studied here. The signal was elusive because it is the integral of noisy momentary evidence, hence stochastic, and undetectable in the firing rates when they are computed as averages over trials. The averages preserve the ramp-like drift component, leaving open the possibility that the averages are composed of other stochastic

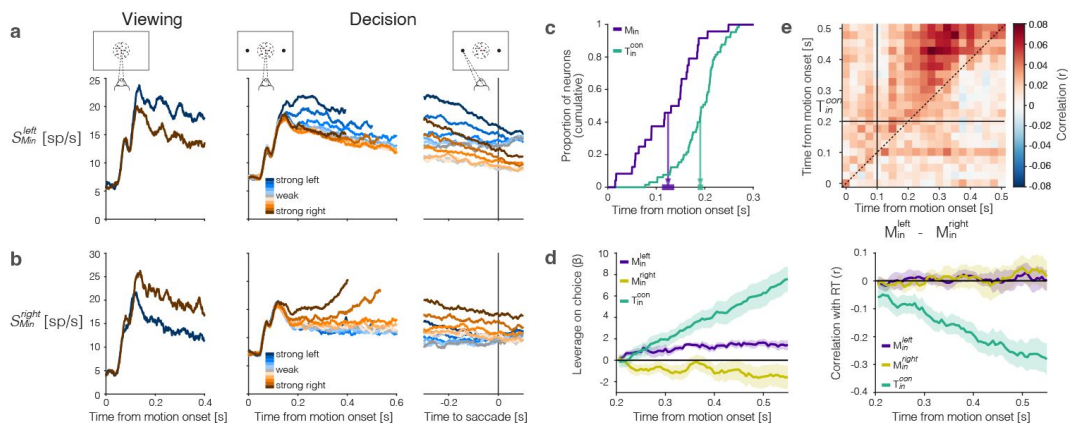


Figure 6.

The representation of momentary evidence in area LIP.

a, Leftward preferring neurons. *left*, Response to strong leftward (blue) and rightward (brown) motion during passive viewing. Traces are averages over neurons and trials. The neurons were selected for analysis based on this task, hence the stronger response to leftward is guaranteed. Note the short-latency visual response to motion onset followed by the direction-selective (DS) response beginning ~ 100 ms after motion onset. *right*, Responses during decision-making, aligned to motion onset and the saccadic response. Response averages are grouped by direction and strength of motion (color legend). The neurons retain the same direction preference during passive viewing and decision-making. The responses are also graded as a function of motion strength. **b**, Rightward preferring neurons. Same conventions as **a**. **c**, Cumulative distribution of the times at which individual neurons start showing evidence-dependent activity. Evidence dependence emerges earlier in M_{in}^{left} and M_{in}^{right} neurons (purple) than in T_{in}^{con} neurons (green). Arrows indicate the mean onset of evidence-dependent activity in each signal. The markers at the end of the arrows show the s.e.m. across neurons. **d**, *left*, Leverage of neural activity on choice for T_{in}^{con} (green), M_{in}^{left} (purple) and M_{in}^{right} (yellow) neurons. *right*, Same as *left*, for the correlation between neural activity and reaction time. The absence of negative correlation is explained by insufficient power (see Methods). **e**, Correlation between the neural representation of motion evidence—the difference in activity of neurons selective for leftward and rightward motion ($M_{in}^{left} - M_{in}^{right}$)—and the neural representation of the decision variable ($S_{T_{in}}^{con}$) across different time points and lags. Horizontal and vertical lines indicate the onset of evidence-dependent activity in each signal. Positive correlations in the upper left triangle indicate that the decision variable at a time point is correlated with earlier activity of the evidence signal.

processes (e.g., Latimer et al., 2015; Cisek et al., 2009). By providing access to populations of neurons in LIP, macaque Neuropixels probes (Trautmann et al., 2023) allowed us to resolve, for the first time, the evolution of LIP activity during a single decision.

The present findings establish that the ramp-like averages arise from drift-diffusion on single trials, and this drift-diffusion signal approximates the decision variable that arbitrates the choice and RT on that trial. We used a variety of strategies to assign a weight to each neuron in the population such that the vector of weights defines a coding direction in neuronal state space. The weighted averages render population firing rate signals on single trials. Our experience is that any method of assigning the weights that captures a known feature of evidence accumulation (or its termination in a saccadic choice) reveals drift-diffusion on single trials, and this also holds for data-driven, hypothesis-free methods such as PCA. This is because the actual dimensionality of the DV is effectively one—a scalar function of time that connects the visual evidence to a saccadic choice (Ganguli et al., 2008). Thus a weighting established by training a decoder at time $t = \tau$ to predict the monkey's choice performs nearly as well when tested at times other than the time the decoder was trained on (i.e., $t \neq \tau$; Fig. 4).

The different strategies for deriving coding directions lead to different weight assignments, but the coding directions are linearly dependent (Fig. 4d). They produce traces, $S(t)$, that are similar (Fig. 4e) and suggestive of drift-diffusion. Traces accompanying trials with the same motion coherence meander and spread apart at a rate similar to diffusion (i.e., standard deviation proportional to $\sqrt{t}$), and they exhibit a pattern of autocorrelation, as a function of time and lag, consistent with diffusion (Fig. 3). The calculations applied in the present study improve upon previous applications (e.g. Churchland et al., 2011; de Lafuente et al., 2015; Shushruth et al., 2018) by incorporating the contribution of the smoothing to the autocorrelations. The departures from theory are explained by the fact that the accumulations are bounded. The upper bound and the fact that spike rates must be non-negative (a *de facto* lower reflecting bound) limits the spread of the single-trial traces.

The single-trial signals, $S_i^x(t)$, approximate the DV that gives rise to the choice and decision time on trial i (Fig. 5 and Fig. S11). Support for this assertion is obtained using a conservative assay, which quantifies the leverage of the first 300 ms of the signal's evolution on decision outcomes—choice and RT—occurring at least 670 ms after motion onset. Naturally, the signals do not explain all the variance of these outcomes. The sample size is limited to N randomly selected, often weakly correlated neurons. The sample size and correlation are especially limiting for the T_{in}^{con} neurons ($E(r) = 0.067 \pm 0.0036$). Control analyses show that the degree of leverage on behavior and mediation of these relationships by later activity is on par with that obtained from simulated, weakly-correlated neurons (Fig. S10, bottom). In addition, because they are identified *post hoc*, many have response fields that barely overlap the choice target. Presumably, that is why their responses are weak compared to previous single-neuron studies in which the choice targets were centered in the response field by the experimenter. Yet even this noisy signal, S_{Tm}^{con} , mediates signals produced by coding directions using the entire population (Fig. 5).

The T_{in}^{con} neurons were the first to be identified as a plausible candidate neural representation of the decision variable, based on firing rate averages (Shadlen and Newsome, 1996; Platt and Glimcher, 1999). This neural type is also representative of the LIP projection to the region of the superior colliculus (SC) that represents the saccadic vector required to center the gaze on the choice target. (Paré and Wurtz, 1998). In a companion study by Stine et al. (2023) we show that the SC is responsible for cessation of integration in LIP. Features of the drift-diffusion signal from the T_{in}^{con} neurons are correlated with bursting events in corresponding populations of T_{in}^{con} neurons in the SC, including the final saccadic burst that ends the decision with a saccade to the contralateral choice target. Stine et al. (2023) also show that inactivation of T_{in}^{con} neurons in the SC has little effect on S_{Tm}^{con} signals in LIP.

Previous studies of LIP using the random dot motion task focused primarily on the T_{in}^{con} neurons (cf. Meister et al., 2013). It was thus unknown whether and how other neurons contribute to the decision process. The Neuropixels probes used in the present study yield a large and unbiased sample of neurons. Many of these neurons have response fields that overlap one of the two choice targets, but the majority have response fields that overlap neither the choice targets nor the random dot motion. Our screening procedures (delayed saccades and passive motion viewing tasks) do not supply a quantitative estimate of their spatial distribution. It is worth noting that neurons with response fields that overlap neither of the two choice targets were assigned nonzero weights by the *What*- and *When*-decoders, and yet, removal of the task-related neurons that represent the choice targets and motion (i.e., T_{in} and M_{in}) decreases decoding accuracy more substantially than removing all but the T_{in} neurons, especially when decoders are not retrained at every time point (Fig. S13). The accuracy the decoder achieves is likely explained by neurons with weak responses that simply failed to meet our criterion for inclusion in the T_{in} and M_{in} categories (e.g., neurons with response fields that barely overlap the choice targets or RDM). Some neurons outside these groups might reflect normalization signals from the T_{in} and M_{in} neurons (Shushruth et al., 2018; Carandini and Heeger, 2012), imbuing broad, decision-related co-variability across the population. It thus seems possible that higher dimensional tasks (e.g., four choices instead of two) could decrease correlations among groups of neurons with different response fields.

The fact that the raw averages from a small number of weakly correlated T_{in} neurons furnish a DV on par with that furnished by the full population underscores the importance of this functional class. The role of the M_{in} neurons is less well understood. Freedman and colleagues described direction selective neurons in LIP, similar to our M_{in} neurons (Freedman and Assad, 2011; Fanini and Assad, 2009; Sarma et al., 2016). They showed that the neurons represent both the direction of motion and the decision in their task. In contrast, we do not observe the evolution of the decision (i.e., DV) by the M_{in} neurons (Fig. 6). The latency of the direction and coherence-dependent signal as well as its dynamics resemble properties of DS neurons in area MT. The delayed correlation between M_{in} and T_{in} responses evokes the intriguing possibility that M_{in} neurons supply the momentary evidence, which is integrated within LIP itself (Zhang et al., 2022). Future experiments that better optimize the yield of M_{in} neurons will be informative, and direct, causal support will require perturbations of functionally identified M_{in} neurons, which is not yet feasible. A natural question is why LIP would contain a copy of the DS signals that are already present in area MT. We suspect it simplifies the routing of momentary evidence from neurons in MT/MST to the appropriate T_{in} neurons. This interpretation leads to the prediction that DS M_{in} neurons would be absent in LIP of monkeys that are naïve to saccadic decisions informed by random dot motion, as has been observed in the SC (Horwitz et al., 2004). Further, when motion is not the feature that informs the saccadic response—for example in a color categorization task (e.g. Kang et al., 2021)—LIP might contain a representation of momentary evidence for color (Toth and Assad, 2002; Sereno and Maunsell, 1998).

The capacity to record from many neurons simultaneously invites characterization of the population in neuronal state space (NSS), in which the activity of each neuron defines a dimension. Often, population activity is confined to a low-dimensional subspace or manifold within the NSS (Vyas et al., 2020). An ever-more-popular viewpoint is that representations within these subspaces are emergent properties of the population—that is, *distributed*, rather than coded *directly* by single neurons—a dichotomy that has its roots in Barlow’s neuron doctrine (as updated in Barlow, 1994). Indeed, it is tempting to conclude that the drift-diffusion signal in LIP is similarly emergent based on our NSS analyses—the identified subspaces (i.e., coding directions) combine neurons with highly diverse activity profiles. In contrast, grouping neurons by the location of their spatial response field reveals a direct coding scheme: T_{in} neurons directly represent the accumulated evidence for making a particular saccade and M_{in} neurons represent the momentary evidence. We argue that this explanation is more parsimonious and, importantly, more principled. Grouping neurons based on spatial selectivity rests on the principle that neurons

with similar RFs have similar projections, which is the basis for topographic maps in the visual and oculomotor systems (Schall, 1995 [↗](#); Silver and Kastner, 2009 [↗](#); Kremkow et al., 2016 [↗](#); Felleman and Van Essen, 1991 [↗](#)). In contrast, there are no principles that guide the grouping of neurons in state space analyses, as the idea is that they may comprise as many dimensions as there are neurons that happen to be sampled by the recording device.

The present finding invites both hope and caution. It may be useful to consider a counterfactual state of the scientific literature that lacks knowledge of the properties of LIP T_{in} neurons—a world without Gnatd and Andersen (1988 [↗](#)) and no knowledge of LIP neurons with spatially selective persistent activity. In this world we have no reason to entertain the hypothesis that decisions would involve neurons that represent the choice targets. We do know about DS neurons in area MT and their causal role in decisions about the direction of random dot motion (Salzman et al., 1992 [↗](#); Fetsch et al., 2018 [↗](#); Ditterich et al., 2003 [↗](#); Liu and Pack, 2017 [↗](#)). We also know that drift-diffusion models explain the choice-response time behavior. Guided by no particular hypothesis, we obtain population neural recordings in the random dot motion task. We do not perform the saccade and passive viewing control experiments. What might we learn from such a dataset? We might apply PCA and/or train a choice decoder or possibly a *When*-decoder. If so, we could discover the drift-diffusion signal and we might also infer that the dimensionality of the signal is low. However, we would not discover the T_{in} neurons without a hypothesis and a test thereof. We might notice that the coding directions that reveal drift-diffusion often render a response at the onset of the choice targets as well as increased activity at the time of saccades to the contralateral choice target. These facts might lead us to hypothesize that the population might contain neurons with visual receptive fields and some relationship to saccadic eye movements. We might then query individual neurons, *post hoc*, for these features, and ask if they render the drift-diffusion signal too. The inferences could then be tested experimentally by including simple delayed saccades in the next experiment. The hope in this counterfactual is that data-driven, hypothesis-free methods can inspire hypotheses about the mechanism. The caution is to avoid the natural tendency to stop before the hypotheses and tests, thus accepting as an endpoint the characterization of population dynamics in high dimensions or a lower dimensional manifold. If LIP is representative, these mathematically accurate characterizations may fail to illuminate the neurobiological parsimony.

Methods

Ethical approval declarations

Two adult male rhesus monkeys (*Macacca mulatta*) were used in the experiments. All training, surgery, and experimental procedures complied with guidelines from the National Institutes of Health and were approved by the Institutional Animal Care and Use Committee at Columbia University. A head post and two recording chambers were implanted under general anaesthesia using sterile surgical procedures (for additional details see So and Shadlen, 2022 [↗](#)). One recording chamber allowed access to area LIP in the right hemisphere. The other was placed on the midline, allowing access to the superior colliculus. Those recordings are described in Stine et al. (2023 [↗](#)). Here we report only on the neural recordings from LIP, focusing on the epoch of decision formation.

Behavioral tasks

The monkeys were trained to interact with visual stimuli presented on a CRT video monitor (Vision Master 1451, Iiyama; viewing distance 57 cm; frame rate 75 Hz) using the psychophysics toolbox (Brainard and Vision, 1997 [↗](#); Pelli, 1997 [↗](#); Kleiner et al., 2007 [↗](#)). Task events were controlled by Rex software (Hays et al., 1982 [↗](#)). The monkeys were trained to control their gaze and make saccadic eye movements to peripheral targets to receive a liquid reward (juice). The

direction of gaze was monitored by an infrared camera (EyeLink 1000; SR Research, Ottawa, Canada; 1 kHz sampling rate). The tasks involve stages separated by random delays, distributed as truncated exponential distributions

$$f(t) = \begin{cases} \frac{\alpha}{\lambda} e^{-\frac{t-t_{\min}}{\lambda}} & t_{\min} \leq t \leq t_{\max} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where $t_{\min}$ and $t_{\max}$ define the range, λ is the time constant, and α is chosen to ensure the total probability is unity. Below, we provide the range ($t_{\min}$ to $t_{\max}$) and the exponential parameter λ for all variable delays. Note that because of truncation, the expectation $\mathbb{E}(t) < t_{\min} + \lambda$.

In the *main task* (**Fig. 1a**) the monkey must decide the net direction of random dot motion and indicate its decision when ready by making a saccadic eye movement to the corresponding choice target. After acquiring a central fixation point and a random delay (0.25–0.7 s, $\lambda = 0.15$), two red choice-targets (diameter 1 dva) appear in the left and right visual fields. The random dot motion is then displayed after a random delay (0.25 – 0.7 s, $\lambda = 0.4$ s) and continues until the monkey breaks fixation. The dots are confined to a circular aperture (diameter 5 dva; degrees visual angle) centered on the fixation point (dot density 16.7 dots·dva⁻²·s⁻¹). The direction and strength of motion are determined pseudorandomly from $\pm\{0, 3.2, 6.4, 12.6, 25.6, 51.2\}\%$ coherence. The sign of the coherence indicates direction (positive for leftward, which is contraversive with respect to the recorded hemisphere). The absolute value of coherence determines the probability that a dot plotted on frame n will be displaced by Δx on frame $n + 3$ ($\Delta t = 40$ ms), as opposed to randomly replaced, where $\Delta x = \frac{5}{75}$ dva, consistent with 5 dva·s⁻¹ speed of apparent motion (see also Roitman and Shadlen, 2002). The monkey is rewarded for making a saccadic eye movement to the appropriate choice target. On trials with 0% coherence motion, either saccadic choice is rewarded with probability $\frac{1}{2}$. Errors are punished by extending the intertrial interval by up to 3 s (see Stine et al., 2023, for additional details). On approximately half of the trials, a 100 ms pulse of weak motion ($\pm 4\%$ coherence) is added to the random dot motion stimulus at a random time (0.1–0.8 s, $\lambda = 0.4$) relative to motion onset (similar to Kiani et al., 2008). Monkey M performed 9684 trials (5 sessions); monkey J performed 8142 trials (3 sessions). The data are also analyzed in a companion paper that focuses on the termination of the decision (Stine et al., 2023).

In the *visually instructed delayed saccade task* (Hikosaka and Wurtz, 1983), one target is displayed at a pseudo-random location in the visual field. After a variable delay (monkey M: 0.4–1.1 s, $\lambda = 0.3$; monkey J: 0.5–1.5 s, $\lambda = 0.2$) the fixation point is extinguished, signalling ‘go’. The monkey is rewarded for making a saccade to within ± 2.5 dva of the location of the target. In a *memory-guided* variant of the task (Gnadt and Andersen, 1988; Funahashi et al., 1989), the target is flashed briefly (200 ms) and the monkey is required to make a saccade to the remembered target location when the fixation point is extinguished. These tasks provide a rough characterization of the neural response fields during the visual, perisaccadic and delay epochs. Neurons are designated $T_{\text{in}}^{\text{con}}$ if they exhibit spatially selective activity at the location of the response target in the visual hemifield contralateral to the recorded hemisphere. This determination is made before analyzing the activity in the random dot motion task. We refer to the unweighted mean firing rate as $S_{T_{\text{in}}}^{\text{con}}$. Neurons are designated $T_{\text{in}}^{\text{ips}}$ if they exhibit spatially selective activity at the location of the response target in the visual hemifield ipsilateral to the recorded hemisphere. These analyses were conducted *post hoc*, after spike sorting.

The *passive motion-viewing task* is identical to the main task, except there are no choice targets and only the strongest motion strength ($\pm 51.2\%$ coherence) is displayed for 500 ms (1 s on a small fraction of trials in session 1). The direction is left or right, determined randomly on each trial ($P_{\text{left}} = \frac{1}{2}$). The monkey is rewarded for maintaining fixation until the random dot motion is extinguished.

Behavioral analyses

We fit a neurally inspired variant of the drift-diffusion model (**Fig. 1c**) to the choice-RT data from each session. The model constructs the decision process as a race between two accumulators: one accumulating evidence for left and against right (e.g., left minus right) and one accumulating evidence for right and against left (e.g., right minus left). The decision is determined by the accumulator that first exceeds its positive decision bound, at which point the decision is terminated. The races are negatively correlated with one another, owing to the common source of noisy evidence. We assume they share half the variance, $\rho = -\sqrt{0.5} \approx -0.71$, but the results are robust to a wide range of reasonable values. The decision bounds are allowed to collapse linearly as a function of time, such that

$$B(t) = B_0 - \alpha t, \quad \alpha \geq 0 \quad (2)$$

We used the method of images [van Den Berg et al. \(2016\)](#); [Shan et al. \(2019\)](#) to compute the probability density of the accumulated evidence for each accumulator (which both start at zero at $t = 0$) as a function of time (t) using a time-step of 1 ms. The decision time distributions rendered by the model were convolved with a Gaussian distribution of the non-decision times, t_{nd} , which combines sensory and motor delays, to generate the predicted RT distributions. The model has six parameters: κ , B_0 , α , μ_{nd} , σ_{nd} , and C_0 , where κ determines the scaling of motion strength to drift rate, C_0 implements bias in units of signed coherence ([Hanks et al., 2011](#)), μ_{nd} is the mean non-decision time and σ_{nd} is its standard deviation (**Table 1**). Additional details about the model and the fitting procedure are described in [van Den Berg et al. \(2016\)](#).

Simulated decision variables

We fit the race model described above to the combined behavioral data across all sessions (separately for each monkey) and used the best-fitting parameters for monkey M (see **Table 1**) to simulate a total of 60,000 trials representing all signed coherences of the motion discrimination task. Each simulated trial yields a time series for two decision variables, one for each accumulator in the race. We assume that the model-derived non-decision time ($t_{nd} = 317$ ms; **Fig. 1b**) comprises visual and motor processing times at the beginning and end of the decision: 200 ms from motion onset to the beginning of evidence integration, and the remaining 117 ms after termination. The latter approximates the variability observed in the saccadic latencies in the delayed saccade task and is simulated using a Normal distribution, $\mathcal{N}(\mu, \sigma)$, where $\mu = 117$ ms and $\sigma = 39$ ms ([Stine et al., 2023](#)). In this variable time period between decision termination and the response (saccade), the simulated DVs were assigned the values they had attained at the start of this epoch. For all analyses that employ these simulations, we use the decision variable for the left-choice accumulator, because the neural recordings were from LIP in the right hemisphere.

Neurophysiology

We used prototype “alpha” version Neuropixels1.0-NHP45 probes (IMEC/HHMI-Janelia) to record the activity of multiple isolated single-units from the ventral subdivision of area LIP (LIP_v; [Lewis and Van Essen 2000](#)). We used anatomical MRI to identify LIP_v and confirmed its physiological hallmarks with single-neuron recordings (Thomas Recording GmbH) before proceeding to multi-neuron recordings. Neuropixels probes enable recording from 384 out of 4416 total electrical contacts distributed along the 45 mm long shank. All data presented here were recorded using the 384 contacts closest to the tip of the probe (Bank 0), spanning 3.84 mm. Reference and ground signals were directly connected to each other and to the monkey’s head post. A total of 1084 neurons were recorded over eight sessions (54–203 neurons per session). (**Table 2**).

Table 1.

Model fit parameters.

κ : scaling of motion strength to drift rate, B_0 : bound height, α : linear urgency component, μ_{nd} : mean of the non-decision time, σ_{nd} : standard deviation of the non-decision time, and C_0 : bias

Parameter	κ	B_0	α	μ_{nd}	σ_{nd}	C_0
monkey M	13.37	1.03	0.4199	0.317	0.039	-0.0144
monkey J	13.72	1.76	1.3591	0.291	0.055	0.0008

Table 2.

Information about individual experimental sessions.

Session	1	2	3	4	5	6	7	8	mean
monkey	M	M	M	M	M	J	J	J	
trials	1797	1696	2256	1859	2076	2449	2799	2894	2228
neurons	191	90	54	140	107	138	161	203	135.5
% T_{in}^{con}	8.9	14.4	16.7	15	17.8	8.7	21.1	13.3	14.5
% T_{in}^{ips}	4.2	5.6	13	13.6	6.5	12.3	0.6	2.0	7.2
% M_{in}^{left}	5.2	5.6	0	5	3.7	2.9	3.7	7.4	4.2
% M_{in}^{right}	1.0	2.2	3.7	1.4	2.8	3.6	2.5	3.0	2.5

The Neuropixels 1.0-NHP45 probe uses a standard Neuropixels 1.0 headstage and is connected via the standard Neuropixels1.0 5m cable to the PCI eXtensions for Instrumentation (PXIe) hardware (PXIe-1071 chassis and PXI-6141 and PXIe-8381 I/O modules, National Instruments). Raw data were acquired using the SpikeGLX software (<http://billkarsh.github.io/SpikeGLX/>), and single-units were identified offline using the Kilosort 2.0 algorithm (Pachitariu et al., 2016; Pachitariu, 2021), followed by manual curation using Phy (<https://github.com/cortex-lab/phy>).

Neural data analysis

The spike times from each neuron are represented as delta functions of discrete time, $s_{i,n}(t)$ on each trial i and each neuron n ($dt = 1$ ms). The weighted sum of these $s_{i,n}(t)$ give rise to the single trial population signals:

$$S_i^x(t) = \sum_n w_n s_{i,n}(t) \quad (3)$$

where the superscript, x , identifies the method or source that establishes the weights—that is, the coding direction in neuronal state space or the neuron type contributing to a pooled average (e.g., T_{in}^{con}). For visualization, the single-trial signals are smoothed by convolution with a truncated Gaussian using the Matlab function, *gausswin* (width = 80 ms, width-factor = 1.5, $\sigma \approx 26$ ms). Unless otherwise specified, all other analyses employ a 50 ms boxcar (rectangular) filter; values plotted at time t include data from $t - 24$ to $t + 25$ ms.

We used several methods to define coding directions in the neuronal state space defined by the population of neurons in each session. For PCA and choice-decoding, we standardized the single trial firing rates for each neuron using the mean and standard deviation of its firing rate at in the epoch $200 \leq t \leq 600$ ms after motion onset. This practice led to the exclusion of two neurons (session 1) that did not produce any spikes in the normalization window. Those neurons were assigned zero weight.

T_{in} neurons. Neurons were classified *post hoc* as T_{in} by visual-inspection of spatial heatmaps of neural activity acquired in the delayed saccade task. We inspected activity in the visual, delay, and perisaccadic epochs of the task. The distribution of target locations was guided by the spatial selectivity of simultaneously recorded neurons in the superior colliculus (SC; see [Stine et al., 2023](#), for details). Briefly, after identifying the location of the SC response fields, we randomly presented saccade targets within this location and seven other, equally spaced locations at the same eccentricity. In monkey J we also included 1–3 additional eccentricities, spanning 5–16 degrees. Neurons were classified as T_{in} if they displayed a clear, spatially-selective response in at least one epoch to one of the two locations occupied by the choice targets in the main task. Neurons that switched their spatial selectivity in different epochs were not classified as T_{in} . The classification was conducted before the analyses of activity in the motion discrimination task. The procedure was meant to mimic those used in earlier single-neuron studies of LIP (e.g., [Roitman and Shadlen, 2002](#)) in which the location of the choice targets was determined online by the qualitative spatial selectivity of the neuron under study. The T_{in}^{con} neurons in the present study were highly selective for either the contralateral or ipsilateral choice target used in the RDM task ($AUC = 0.89 \pm 0.01$; $p < 0.05$ for 97% of neurons, Wilcoxon rank sum test). Given the sparse sampling of saccade target locations, we are unable to supply a quantitative estimate of the center and spatial extent of the RFs. We next describe the methods to establish the coding directions.

Ramp direction

We applied linear regression to generate a signal that best approximates a linear ramp, on each trial, i , that terminates with a saccade to the choice-target contralateral to the hemisphere of the LIP recordings. The ramps are defined in the epoch spanning the decision time: each ramp begins at $f_i(t_0) = -1$, where $t_0 = 0.2$ s after motion onset, and ends at $f_i(t_1) = 1$, where $t_1 = t_{sac} - 0.05$ s (i.e., 50

ms before saccade initiation). The ramps are sampled every 25 ms and concatenated using all eligible trials to construct a long saw-tooth function (see **Fig. S2**). The regression solves for the weights assigned to each neuron such that the weighted sum of the activity of all neurons best approximates the saw-tooth. We constructed a time series of standardized neural activity, sampled identically to the saw-tooth. The spike times from each neuron are represented as delta functions (rasters) and convolved with a non-causal 25 ms boxcar filter. The mean and standard deviation of all sampled values of activity were used to standardize the activity for each neuron (*i.e.*, Z-transform). The coefficients derived from the regression establish the vector of weights that define S^{ramp} . The algorithm ensures that the population signal $S^{\text{ramp}}(t)$, but not necessarily individual neurons, have amplitudes ranging from approximately -1 to 1.

We employed a lasso linear regression with $\lambda = 0.005$. The vector of weights assigned across the neurons defines a direction in neuronal state space, S^{ramp} , which we use to render the signal $S^{\text{ramp}}(t)$ on single trials by projecting the data onto this direction. To determine the effect of the regularization term in the lasso regression, we recomputed single-trial signals using standard linear regression, without regularization. We then calculated the Pearson correlation between single-trial traces generated by projecting neural data onto the two coding directions (*i.e.*, with and without regularization). The high correlation between single trial traces (mean $r = 0.99$, across sessions) indicates that the findings are not a result of the regularization applied. Here and elsewhere we compute the mean r using the Fisher-z transform, such that

$$\bar{r} = Z^{\text{inv}} \left(\mathbb{E} [Z(r_k)] \right) \quad (4)$$

where Z^{inv} is the inverse Fisher-z.

Principal Component Analysis (PCA)

We applied standard PCA to the firing rate averages for each neuron using all trials sharing the same signed motion coherence in the shorter of two epochs: 200 ms to either 600 ms after motion onset or 100 ms before the median RT for the signed coherence, whichever produces the shorter interval. The results of the PCA indicate that the dimensionality of the data is low, but greater than one. The participation ratio is 4.4 ± 1.3 (Mazzucato et al., 2016; Gao et al., 2017) and the first 3 PCs explain 67.1 ± 3.1 % of the variance on average (both mean $\pm$ s.e.m. across sessions). As in all other analyses of neural activity aligned to motion onset, we exclude data in the 100 ms epoch ending at saccade initiation on each trial. We projected the neural data onto the first PC to generate the signal S^{PC1} .

Choice decoder

For each experimental session, we trained logistic choice decoders with lasso regularization ($\lambda = 0.01$) on the population activity in 50 ms time bins spanning the first 500 ms after motion onset and the 300 ms epoch ending at saccade initiation, respectively. Each of the decoders was trained on the even-numbered trials. Decoder accuracy was cross-validated using the activity of held-out, odd trials at the same time point (**Fig. 4a**). For the time bins aligned to motion onset, we also assessed the accuracy of the decoders trained on each of the time bins to predict the choice on time bins on which they were not trained (**Fig. 4b**; King and Dehaene, 2014). We use the decoder trained on the bin centered on $t = 450$ ms to define the *What* coding direction. We refer to this as the *fixed training-time* decoder to distinguish it from the standard machine-learning decoder, which assigns a potentially distinct vector of weights at each time point. We applied a similar analysis to simulated data (see *Simulated decision variables*) to generate the black curve in **Fig. 4a**. Assuming a stochastic drift-diffusion process giving rise to choice and response times, the exercise establishes a rough upper bound on decoder accuracy, were the actual drift-diffusion process known precisely.

When decoder

This decoder is trained to predict whether a saccade to the left (contralateral) choice target will occur within the next 150 ms. We applied logistic regression with lasso regularization ($\lambda = 0.01$) to spike counts from each neuron in discrete bins of 25 ms, from 200 ms before motion onset to 50 ms before the saccade. We used only trials ending in a left choice (including errors) and trained the decoder on the even numbered half of those trials. The concatenation of these trials forms a sequence of step functions which are set to 1 if a saccade occurred within 150 ms of the start of the 25 ms time bin and 0 otherwise (**Fig. S6** [↗](#)).

The spike counts were also concatenated across these trials to construct column vectors (one per neuron) that match the vector of concatenated step functions. These concatenated vectors, one per neuron, plus an offset (β_0), serve as the independent variables of the regression model (one β term per neuron). The proportion of β weights equal to zero, controlled by the lasso parameter, λ , was 0.8 ± 0.02 across sessions. The weights define the S^{When} coding direction, which yields single-trial signals, $S^{\text{When}}(t)$. The When-decoder signal is $S^{\text{When}}(t) + \beta_0$. We validated the When-decoder by computing the area under an ROC (AUC) using the held-out (odd-numbered) trials ending in left choices (mean AUC over all time points and sessions: 0.84 ± 0.024 , mean $\pm$ s.e.).

Our motivation, however, was to ascertain whether the *When* coding direction also predicts the monkey's choices on *all trials*—that is, to perform as a *What* decoder. To this end, we predicted the choice using the sign of the detrended $S^{\text{When}}(t)$, formed by subtracting the average of the signal using all trials:

$$S_i^*(t) = S_i^{\text{When}}(t) - \langle S_i^{\text{When}}(t) \rangle_i \quad (5)$$

where $\langle \dots \rangle_i$ denotes expectation across all trials contributing values at time t . The choice accuracy is

$$A_i(t) = \begin{cases} 1 & S_i^*(t) > 0 \text{ \& choice}_i = \text{left} \\ 1 & S_i^*(t) \leq 0 \text{ \& choice}_i = \text{right} \\ 0 & \text{otherwise} \end{cases}$$

The green trace in **Fig. 4a** [↗](#) shows $\langle A_i(t) \rangle_i$.

Aggregation of data across experimental sessions

To combine single trial data across sessions (e.g. $S^x(t)$), we first normalize activity within each session as follows. Using all trials ending in the same choice, we construct the trial-averaged activity aligned to both motion onset ($0 \leq t_{\text{motion}} \leq 0.6$ s) and saccade onset ($-0.6 \leq t_{\text{sacc}} \leq 0$ s). This produces four traces. The minimum and maximum values (a_{min} and a_{max}) over all four traces establish the range, zero to one, of the normalized signal:

$$\hat{s}^x(i, t) = \frac{s^x(i, t) - a_{\text{min}}}{a_{\text{max}} - a_{\text{min}}} \quad (6)$$

where lower-case $\hat{s}^x(i, t)$ is the normalized signal across trials i and time t in an individual session.

Cosine similarity

We computed the cosine similarity between the weight vectors that define coding directions S^{ramp} , S^{PC1} , $S^{\text{con}}_{\text{Tin}}$, S^{What} and S^{When} . Mean cosine similarities are portrayed in the heatmap in **Fig. 4d** and also in **Fig. S8**, *top*, where they are accompanied by error bars. We evaluated the null hypothesis that the mean cosine similarity is ≤ 0 with t-tests. We also performed two control analyses that deploy random coding directions in neuronal state space. For each of the original coding directions we obtained 1000 random coding directions as random permutations of the weight assignments. The cosine similarities between pairs of such random directions in state space are shown in **Fig. S8**, *top*. The cumulative distribution of cosine similarities under permutation supports p -values less than 1/1000. In a second control analysis, we used random unit vectors as random coding directions (Normal distribution with mean 0 and scaled to unit length).

Similarity of single-trial signals

We calculated Pearson correlation to quantify the similarity of the signals generated by pairs of coding directions, x and y . For each trial, i , the detrended signals, $\tilde{S}_i^x$ and $\tilde{S}_i^y$, provide ordered pairs, $\{\tilde{S}_i^x(t_j), \tilde{S}_i^y(t_j)\}$ where j indexes successive 50 ms bins between 200 ms after motion onset and 100 ms before saccade initiation. We excluded trials comprising less than four such bins. Each trial gives rise to a correlation coefficient, r_i . We report the mean r using **Eq. 4**. The r -values for comparisons across all pairs of coding directions are summarized in **Fig. 4e**, and variability across sessions is portrayed in **Fig. S8**, *bottom*. For each pair of CDs and session, we evaluated the null hypothesis, $\mathcal{H}_0: \bar{r} \leq 0$ (t-test).

We also performed two control analyses that deploy random (CDs) in neuronal state space. These analyses control for the possibility that the correlations observed in the signals are explained by pairwise correlations between the neurons, regardless of the signals produced by the weighted sums. (1) We generated sets of single-trial traces $S^{\text{rand}}(t)$ by projecting the neural responses onto random CDs, defined by permuting the weights of each coding direction (S^{ramp} , S^{PC1} , $S^{\text{con}}_{\text{Tin}}$, S^{What} and S^{When}). For each pair of CDs, we compute within-trial correlations between ordered pairs of trials using the same method applied to the original signals. We repeat this process for a total of 1,000 random permutations per pair of CDs, per session. (2) We sample a pair of random weight vectors from standard Normal distributions. Each weight vector has a dimension equal to the number of recorded neurons in the session. The weight vectors are normalized to sum to 1. We generate random CDs using these weights and compute within-trial correlations using the same method applied to the original signals. We repeat this process 1,000 times per session. For both analyses, we evaluated the null hypothesis that the observed correlations are not greater than those produced by the random projections (t-tests using the Fisher-z transformed correlations). Mean $\pm$ stdev of the mean r -values between ordered pairs for both control analyses are summarized in **Fig. S8**, *bottom*.

Leverage of single-trial activity on behavior

The leverage of single-trial signals, $S^x(t)$, on choice and RT was assessed using the earliest 300 ms epoch of putative integration ($0.2 < t < 0.5$ s from motion onset), restricting analyses to trials with RTs outside this range ($0.67 < \text{RT} < 2$ s). The single-trial signals are smoothed with a 50 ms boxcar filter and detrended by subtracting the mean $S(t)$ for trials sharing the same motion strength and direction (*i.e.*, signed coherence). The reaction times are also expressed as residuals relative to the mean RT, using trials sharing the same signed coherence and choice. We include trials with $|\text{coh}| \leq 0.064$ that result in choices of the left response target in this analysis. Including trials of $|\text{coh}| \leq 0.128$ produced comparable results. The leverage on RT is the Pearson correlation between the residual signals $\tilde{S}(t)$ at each time $0.2 < t \leq 0.5$, and $\tilde{\text{RT}}$ on that trial, where the tilde indicates residual. Correlations were computed per session and then averaged across sessions (**Eq. 4**). We

also show the correlation at $t=0.55$ s, using the ordered pairs, $\{\tilde{S}(t = 0.55), \tilde{RT}\}$. We quantify mediation of the leverage of earlier samples by the later sample of S^x by computing partial correlations $\{\tilde{S}(t), \tilde{RT} \mid \tilde{S}(t = 0.55)\}$, also notated $R_{\tilde{S}(t), \tilde{RT} \mid \tilde{S}(0.55)}$ in **Fig. 5**. We show this mediation at all time points. We also report a mediation statistic (ξ^{RT}) using the time point 200 ms after the beginning of putative integration (i.e., $S(t = 0.4)$):

$$\xi^{RT} = 1 - \frac{R_{\tilde{S}(0.4), \tilde{RT} \mid \tilde{S}(0.55)}^2}{R_{\tilde{S}(0.4), \tilde{RT}}^2} \quad (7)$$

The rationale for using the 400 ms time point is (i) to allow the process to have achieved enough leverage on RT so that a reduction is meaningful and (ii) to preserve a substantial gap between this time and the sample at $t = 0.55$ s (e.g., to avoid autocorrelations imposed by smoothing). The rare cases in which there was no negative correlation between $S(t = 0.4)$ and RT were excluded from this summary statistic, ξ^{RT} , because no mediation is possible (Session 2: S^{ramp} & S^{PC1}). When combining values of ξ^{RT} across sessions, we rectify any $R_{\tilde{S}(t), \tilde{RT} \mid \tilde{S}(0.55)} > 0$ to 0. This occurs rarely when the mediated correlation is near zero, typically at early times.

We compute the leverage on choice, ξ^{Ch} , using trials with $|\text{coh}| \leq 0.064$ and the same time points as for ξ^{RT} . Instead of an R-squared measure, we based ξ^{Ch} on coefficients derived from logistic regression:

$$P^{\text{con}}(t) = \frac{1}{1 + e^{-(\beta_0^{\text{coh}} + \beta_1(t)S(t))}}, \quad 0.2 \leq t \leq 0.5. \quad (8)$$

where β_0^{coh} is a set of constants that accounts for the proportion of contralateral choices at each signed motion strength and $\beta_1(t)$ is the simple leverage of $S(t)$ on choice, analogous to simple correlation. The regression analysis was performed separately for each session. The coefficients $\beta_1(t)$ were divided by their standard error and then averaged across sessions. This normalization step was implemented to control for potential variation in the magnitude of $S(t)$ (and therefore of $\beta_1(t)$) across sessions. Analogous to partial correlation, we include the later time point and fit

$$P^{\text{con}}(t) = \frac{1}{1 + e^{-(\beta_0^{\text{coh}} + \beta_1^*(t)S(t) + \beta_2 S(0.55))}} \quad (9)$$

where $\beta_1^*(t)$ is the amount of leverage at time t given the $S(0.55)$. The regression coefficients $\beta_1^*(t)$ were averaged across sessions after dividing them by the standard error of the $\beta_1(t)$ coefficients obtained from **Eq. 8**. That is, the same normalization factors were used for the mediated and unmediated leverage. The summary statistic for choice mediation is defined by

$$\xi^{\text{Ch}}(t) = \begin{cases} 1 & \beta_1^*(t) \leq 0 \\ 1 - \frac{\beta_1^*(t)}{\beta_1(t)} & \beta_1^*(t) > 0 \\ \text{undefined} & \beta_1(t) \leq 0 \text{ (i.e., mediation is not possible)} \end{cases} \quad (10)$$

For both types of mediation, we also test whether the earlier $S(t)$ is mediated by the $T_{\text{in}}^{\text{con}}$ neurons by substituting $S_{T_{\text{in}}^{\text{con}}}(0.55)$ for $S(0.55)$ in **Eq. 9** and in the expression for partial correlations.

Because the mediation statistics, ξ^{RT} and ξ^{Ch} , are, by their definition, non-negative, we assess statistical significance by bootstrapping. For each session, we construct 1000 surrogate data sets equal in size to the original data, by sampling with replacement. The standard deviation of the leverage and mediation values at each time approximates the standard error. We compare the distribution of the mediation statistics, ζ , to their distribution under the null hypothesis, $\mathcal{H}_0$, that the values arise by chance, instantiated by breaking the correspondence with the trial giving rise

to the later sample, $S^X(t = 0.55s)$. The permutation maintains correspondence in signed motion coherence. We compare the distributions of mediation from the bootstrap and $\mathcal{H}_0$ using the Wilcoxin Ranked Sum test.

To test whether the observed leverage of neural activity on choice and RT is achieved by projections onto arbitrary coding directions, we generated random weight vectors by permuting the weights associated with the first PC for each session. We projected activity onto this random coding direction, applied the mediation analyses described above to this signal, and repeated this process 1000 times to produce a null distribution at each time point. The reported p-values represent the probability that the observed leverage was generated from this null distribution.

We performed a similar analysis to test whether the observed leverage depends on the trial-to-trial correspondence between neural activity and behavior. Here, the null distribution at each time point was generated by randomly permuting the trial indices associated with the neural activity and those associated with the behavioral measures.

Finally, we used the simulated data from the racing accumulator model to test the degree of leverage and mediation expected had we known the ground-truth decision variable on each trial (**Fig. S10**). Because the simulated DV is noiseless (and the process is Markovian), the mediation is expected to be complete for all time points tested in the analyses. We therefore took two steps to make the simulated data more comparable to neural data: (i) We sub-sampled the simulated data to match the number of trials in each session. (ii) We generated N noisy instantiations of the signal for each of the sub-sampled, simulated trials, where N is the number of T_{in}^{con} neurons in each session. The added noise is independent across time points and weakly correlated across all $N \times (N - 1)/2$ neuron pairs ($r \approx 0.09$). We then applied the mediation analyses to the mean of these signals and repeated this process 1000 times.

We performed three control analyses to determine whether the results of the mediation analysis were specific, meaningful and comparable to results obtained for the DV of a race model. To showcase that the results are specific, we generated random weight vectors by permuting the weights of the PC1 coding direction. We repeated this procedure 1000 times per session and projected the data along these directions in neuronal state space to generate S^{rand} . We then computed the mediation analyses detailed above on these signals and determined significance by comparing the leverage on choice and correlation with RT of S^{ramp} , S^{PC1} , and S_{Tin}^{con} to the null distributions of S^{rand} at each time point.

To estimate an upper limit for the degree of possible leverage and mediation, we simulated 60,000 trials using the race model that best fits the behavioral data of monkey M (see *Simulated decision variables*). For any noise-free representation of a Markovian integration process, the leverage of an early sample of the DV on behavior would be mediated completely by later activity as the latter sample by definition encompasses all variability captured by the earlier sample. We, therefore, took two steps to make the simulated decision variables more comparable to real neural data. (1) For each session, we first subsampled the simulated data to match the each session. (2) To evaluate a DV approximated from the activity of n T_{in}^{con} neurons per session rather than the true DV represented by the entire population, we then generated n noisy instantiations of the signal for each simulated trial. The added noise independent across time points and weakly correlated across neurons ($r \approx 0.09$). We then computed the measured DV S^{sim} as the mean activity of these n simulated neurons. We repeated this procedure 1000 times per session and **Fig. S10**, bottom displays the mean and standard deviation across permutations of the leverage of $S^{sim}(t)$ on behavior. The simulation results highlight that we would not expect the mediation of the leverage on behavior by a later sample to be complete (blue traces in **Fig. 5** and **Fig. S10** not aligned with zero).

Noise correlation between neurons

The mean pairwise correlation between T_{in}^{con} neurons, reported in Results, is based on all pairs of simultaneously recorded T_{in}^{con} neurons in each session and all trials with $RT > 0.5$ s. For each neuron and each trial, we compute the time-averaged activity over the epoch $0.2 \leq t \leq 0.4$. These scalar values are converted to residuals by subtracting the mean (for each neuron) across all trials sharing the same signed motion coherence. The residuals from all eligible trials are concatenated for each neuron to support the calculation of $N \times (N - 1)$ Pearson r values, where N is the number of T_{in}^{con} neurons in the session. The mean correlation for all pairs of T_{in}^{con} neurons across all sessions is computed using Eq. 4 [↗](#).

Direction selective neurons

We identified direction-selective (DS) neurons (M_{in} neurons) using the passive motion-viewing task (described above). We classified a neuron as M_{in} if it achieved two criterion. The first criterion is a short-latency response to the onset of random dot motion, which we defined as a 5-fold increase in firing rate relative to baseline in the first 80 ms following motion onset or a steeper increase in activity in the 80 ms after than the 200 ms before motion onset. The second criterion was direction selectivity. We calculated the area under the ROC (AUC) comparing leftward versus rightward for two separate epochs: (i) $0.15 \leq t < 0.3$ s and (ii) $0.3 \leq t < 0.5$ s. Neurons were determined to be DS if the AUC in either epoch exceeded 0.6. We excluded one neuron from this analysis because it switched its direction preference in the two epochs. We also excluded neurons that had previously been classified as T_{in} .

Latency analysis

We estimated the latency of direction-selective responses using the CUSUM method (Ellaway, 1978 [↗](#); Lortelje et al., 2015 [↗](#)). We employed a receiver operating characteristic (ROC) analysis to estimate the selectivity of each M_{in} neuron to motion direction. The AUC reflects the separation of the distributions of spike counts (100 to 400 ms after motion onset) on single trials of leftward and rightward motion, respectively. We included only correct trials with response times greater than 450 ms and motion strengths above 10% coherence. For each neuron with $AUC > 0.6$, we computed the difference in spike counts (25 ms bins) between correct trials featuring leftward and rightward motion. Subsequently, we accumulated these differences over time, following the CUSUM method. The resulting difference is approximately zero before the onset of direction selectivity and then either increases or decreases monotonically, depending on the preferred motion direction. To identify the transition between these two regimes, we fit a dog leg function to the cumulative sum of spikes: a flat line starting at $t_0 = 0$ followed by a linearly increasing component beginning at $t_1 > t_0$. The time of the end of the flat portion (between 0 and 500 ms from motion onset) of the fit was taken as the latency. Estimating latencies based on cumulative sums of spikes helps mitigate the effect of neuronal noise. The fitting step reduces the effect of the number of trials on latency estimates compared to traditional methods that rely on t-tests in moving windows.

Correlations between M_{in} and T_{in}^{con}

The analysis of the correlations shown in Fig. 6e [↗](#) is based on the spike counts of the M_{in}^{left} , M_{in}^{right} , and T_{in}^{con} neurons calculated in 25 ms windows. We computed residuals by subtracting from each trial and time bin its average over trials of the same signed coherence. The spike count residuals were then z-scored independently for each time bin and session. Trials from different sessions were concatenated, and the baseline activity—last 100 ms before motion onset—was subtracted from each trial. We refer to the resulting signals as $\tilde{S}_{Min}^{left}$, $\tilde{S}_{Min}^{right}$, and $\tilde{S}_{Tin}^{con}$. Trials with response time

less than 0.55 s were discarded, and the correlations between the difference, $\tilde{S}_{\text{Min}}^{\text{left}} - \tilde{S}_{\text{Min}}^{\text{right}}$, and $\tilde{S}_{\text{Tin}}^{\text{con}}$ were calculated for all pairs of time steps between 0 and 500 milliseconds (Fig. 6e). Statistical significance was assessed using permutation tests, as follows. Two regions of interest (ROI) were defined based on the time from stimulus onset for the M_{in} (x) and T_{in} (y) dimensions. The first region of interest, ROI₁, is characterized by $t_x > 100$, $t_y > 200$, and $t_y > t_x$. According to our hypothesis that the M_{in} neurons represent the momentary evidence integrated by $T_{\text{in}}^{\text{con}}$ neurons, we anticipated high correlations in this region. The second region of interest, ROI₂, is defined by $t_x > 100$, $t_y > 200$, and $t_y < t_x$. If, contrary to our hypothesis, M_{in} and T_{in} signals were influencing each other bidirectionally, we would expect high correlations in this region. We calculated the difference in correlations between these two groups, $(\rho_{\text{ROI1}}) - (\rho_{\text{ROI2}})$, where the expectation is over the time bins within each region of interest. This difference was compared to those obtained after randomly shuffling the order of the trials for one of the dimensions before calculating the pairwise correlations ($N_{\text{shuffles}} = 200$). We assess significance with a z-test given the mean and standard deviation of the values obtained under shuffling. The analysis was repeated with an alternative ROI₂ defined by $t_x < 100$ and $t_y < 200$, representing the times before direction selectivity is present in at least one of the two dimensions.

Correlations between M_{in} signals and behavior

To assess the leverage of M_{in} signals on choice and RT (Fig. 6d) we performed the same logistic regression and pairwise correlation analyses as in Fig. 5, substituting the $M_{\text{in}}^{\text{left}}$ and $M_{\text{in}}^{\text{right}}$ for S^X . The leverage on choice is not mediated by a later sample of either M_{in} signal ($\xi^{\text{Ch}} \leq 9.6\%$; not shown), and there is negligible leverage on RT to mediate. We suspect the failure to detect leverage of M_{in} is explained by a lack of power, owing to the focus on long RT trials, narrow sample windows (50 ms boxcar), and the small number of $M_{\text{in}}^{\text{left}}$ and $M_{\text{in}}^{\text{right}}$ neurons. We support this suspicion with a simpler correlation analysis, using the difference of the M_{in} signals (standardized as in the previous paragraph):

$$\psi_k = \text{mean} \left(\tilde{S}_{\text{Min}}^{\text{left}}(t) - \tilde{S}_{\text{Min}}^{\text{right}}(t) \right) \quad (11)$$

on the interval $0.1 \leq t \leq 0.4$ s from motion onset, on each trial, k , including trials with contraversive choices and $\text{RT} \geq 500$ ms. We calculated the Pearson correlation coefficient between ψ_k and RT. Response times were z-scored independently for each signed motion strength and session. We evaluated the null hypothesis that the correlation coefficient is non-negative. The reported p-value is based on a one-tailed t-statistic.

Variance and autocorrelation of smoothed diffusion signals

The analyses in Fig. 3 compare the variance and autocorrelation of the single-trial signals, $S^{\text{ramp}}(t)$, to those expected from unbounded drift-diffusion. To mitigate the effect of the bound, we focus on the earliest epoch of putative integration (200 to 506 ms after motion onset; six 51 ms counting windows) and the weakest motion strengths ($|\text{coh}| \leq 3.2\%$). The single trial signals are detrended by the mean across trials sharing the same signed motion coherence and baseline corrected by subtraction of $S_i^{\text{ramp}}(t = 0.2)$ from all time points on each trial i .

The variance as a function of time and the autocorrelation as a function of time and lag are well specified for the cumulative sum of discrete *iid* random samples, but the autocorrelation is affected by the boxcar filter we applied to render the signals. We incorporated the correction in our characterization of unbounded diffusion. The derivation is summarized in Appendix 1, and we provide Matlab code in the GitHub repository. The theoretical values shown in Fig. 3 assume a 1 kHz sampling rate and standard Wiener process (i.e., samples drawn from a normal distribution with $\{\mu = 0, \sigma = \sqrt{dt}\}$). The evolution of variance would be a line from 0 to 1 over the first second

of integration. The key prediction, shown in **Fig. 3a** is that the variance of mean single-trial signals, $S_i(t)$ over the epoch 26 ± 25 ms should double in the epoch $(26 + 51) \pm 25$ ms, and triple in the epoch $(26 + 2 \times 51) \pm 25$ ms, and so on for each successive non-overlapping running mean. We therefore use arbitrary units, normalized to the measured variance of the first point. We do not know the variance of the drift-diffusion signal that $S(t)$ is thought to approximate, but we assume it can be decomposed—by the law of total variance—to a component given by drift-diffusion and components associated with spiking and other nuisance factors. We therefore search for a scalar non-negative factor $\varphi \leq 1$ that multiplies all terms in the diagonal of the empirical covariance matrix (i.e., the variance) before normalizing to produce the autocorrelation matrix. We search for the value of φ that minimizes the sum of squares between Fisher-z transformed correlation coefficients in the theoretical and empirical autocorrelation matrices (**Fig. 3b,c**). Standard errors of the variance and autocorrelations in (**Fig. 3a & c**) are estimated by a bootstrap procedure respecting the composition of motion strength and direction (the s.e. is standard deviation of each variance and autocorrelation term across 500 repetitions of the procedure).

Appendix

We consider a discrete time (sampling interval dt) Wiener process with independent random increments ϵ_k on time step k that are zero-mean noise with variance $\sigma^2(\epsilon_k) = dt$ (i.e. unit variance per second). The accumulated evidence (i.e. decision variable, DV) on time step p is

$$DV_p = \sum_{k=1}^p \epsilon_k \quad (12)$$

For such a Wiener process,

$$\text{Cov}(DV_p, DV_q) = \text{Cov}\left(\sum_{j=1}^p \epsilon_j, \sum_{k=1}^q \epsilon_k\right) = \sum_{j=1}^p \sum_{k=1}^q \text{Cov}(\epsilon_j, \epsilon_k)$$

As the increments ϵ_k are independent across time, $\text{Cov}(\epsilon_j, \epsilon_k)$ is 0 for $j \neq k$ and dt for $j = k$

$$\text{Cov}(DV_p, DV_q) = \sum_{j=1}^{\min(p,q)} \text{Cov}(\epsilon_j, \epsilon_j) = \min(p, q)dt.$$

We define the mean DV over a window of $\pm n$ points as

$$MDV_p^n = \frac{1}{2n+1} \sum_{j=-n}^n DV_{p+j} \quad (13)$$

We consider two time points $p < q$ with window n such that there is no overlap and hence $p + n < q - n$

$$\begin{aligned}\text{Cov}(MDV_p^n, MDV_q^n) &= \text{Cov}\left(\frac{1}{2n+1} \sum_{j=-n}^n DV_{p+j}, \frac{1}{2n+1} \sum_{k=-n}^n DV_{q+k}\right) \\ &= \frac{1}{(2n+1)^2} \sum_{j=-n}^n \sum_{k=-n}^n \text{Cov}(DV_{p+j}, DV_{q+k}) \\ &= \frac{dt}{(2n+1)^2} \sum_{j=-n}^n \sum_{k=-n}^n \min(p+j, q+k) \\ &= \frac{dt}{(2n+1)^2} \sum_{j=-n}^n \sum_{k=-n}^n p+j \\ &= \frac{dt}{(2n+1)^2} (2n+1)^2 p = p \cdot dt\end{aligned}$$

$$\begin{aligned}\text{Var}(MDV_p^n) &= \text{Var}\left(\frac{1}{2n+1} \sum_{j=-n}^n DV_{p+j}\right) \\ &= \frac{1}{(2n+1)^2} \sum_{j=-n}^n \sum_{k=-n}^n \text{Cov}(DV_{p+j}, DV_{p+k}) \\ &= \frac{1}{(2n+1)^2} \sum_{j=1}^{2n+1} \sum_{k=1}^{2n+1} \text{Cov}(DV_{p-n-1+j}, DV_{p-n-1+k}) \\ &= \frac{dt}{(2n+1)^2} \sum_{j=1}^{2n+1} \sum_{k=1}^{2n+1} \min(p-n-1+j, p-n-1+k) \\ &= \frac{dt}{(2n+1)^2} \left[(2n+1)^2 (p-n-1) + \sum_{j=1}^{2n+1} \sum_{k=1}^{2n+1} \min(j, k) \right]\end{aligned}$$

$$\text{Given that } \sum_{j=1}^n \sum_{k=1}^n \min(j, k) = \sum_{j=1}^n j^2 = n(n+1)(2n+1)/6$$

$$\begin{aligned}\text{Var}(MDV_p^n) &= \frac{dt}{(2n+1)^2} [(2n+1)^2 (p-n-1) + (2n+1)(2n+2)(4n+3)/6] \\ &= \frac{3p+6np-2n-2n^2}{3(2n+1)} dt\end{aligned}$$

Therefore the correlation

$$\begin{aligned}\text{Corr}(MDV_p^n, MDV_q^n) &= \frac{\text{Cov}(MDV_p^n, MDV_q^n)}{\sqrt{\text{Var}(MDV_p^n) \text{Var}(MDV_q^n)}} \\ &= \frac{3p(2n+1)}{\sqrt{(3p+6np-2n-2n^2)(3q+6nq-2n-2n^2)}}\end{aligned}$$

In contrast for the point estimates at p and q

$$\text{Corr}(DV_p, DV_q) = \sqrt{\frac{p}{q}}$$

It is useful to re-express the above two equations in terms of actual time t_p and t_q and window size t_n . Substituting for p , q , and n with t_p/dt , t_q/dt and t_n/dt

$$\begin{aligned}\text{Corr}(MDV_{t_p}^{t_n}, MDV_{t_q}^{t_n}) &= \frac{3t_p(2t_n + dt)}{\sqrt{(3t_p dt + 6t_n t_p - 2t_n dt - 2t_n^2)(3t_q dt + 6t_n t_q - 2t_n dt - 2t_n^2)}} \\ &= \frac{3t_p(2t_n + dt)}{\sqrt{((3t_p - 2t_n)dt + 6t_n t_p - 2t_n^2)((3t_q - 2t_n)dt + 6t_n t_q - 2t_n^2)}} \\ \text{Corr}(DV_{t_p}, DV_{t_q}) &= \sqrt{\frac{t_p}{t_q}}\end{aligned}$$

Acknowledgements

We thank Shushruth, NaYoung So, and David Gruskin for comments on the manuscript, Cornel Duhaney and Brian Madeira for their assistance in the planning and execution of surgeries, animal training and general support, and we thank Columbia University's ICM for the quality of care they provide for our animals, especially during the pandemic and lockdown. We would further like to thank Tanya Tabachnik and her team at the Zuckerman Institute Advanced Instrumentation Core and Tim Harris, Wei-lung Sun, Jennifer Colonell, and Bill Karsh at HHMI Janelia for their continued support with Neuropixels1.0-NHP45 probes development and testing. This research was supported by the Howard Hughes Medical Institute; an R01 grant from the NIH Brain Initiative (M.N.S., R01NS113113); a T32 and F31 grant from the National Eye Institute (G.M.S., T32 EY013933, F31 EY032791); the Grossman center; and the Brain and Behavior Research Foundation.

Author contributions

Natalie A Steinemann and Gabriel M Stine contributed equally to this work. Ariel Zylberberg, Daniel M Wolpert, and Michael N Shadlen contributed equally to this work.

G.M.S. and M.N.S. designed the experiment. N.A.S., G.M.S., and E.M.T. implemented the experimental setup. G.M.S. collected the data. N.A.S., G.M.S., A.Z., D.M.W. and M.N.S. analyzed the data. All authors wrote the manuscript.

Data availability statement

Data and code are available upon request.

Extended Data

Figure S1.

Effect of motion pulses on behavior (adapted from Stine et al., 2023 [link](#)).

a, Choice (bottom) and mean reaction time (top) as a function of motion strength (combined data from the two monkeys; otherwise same conventions as **Fig. 1b** [link](#)). The two traces show trials in which a leftward (grey) and rightward (black) motion pulse occurred during motion viewing. The pulses (100 ms) had a biasing effect on RT and choice equivalent to shifting the functions left or right by $\pm 1.4\%$ coh ($p < 0.001$, likelihood ratio test). **b**, Effect of motion pulses on choices as a function of time from the response. Pulses had a persistent effect on choices, consistent with temporal integration of motion evidence. Shading is ± 1 s.e.

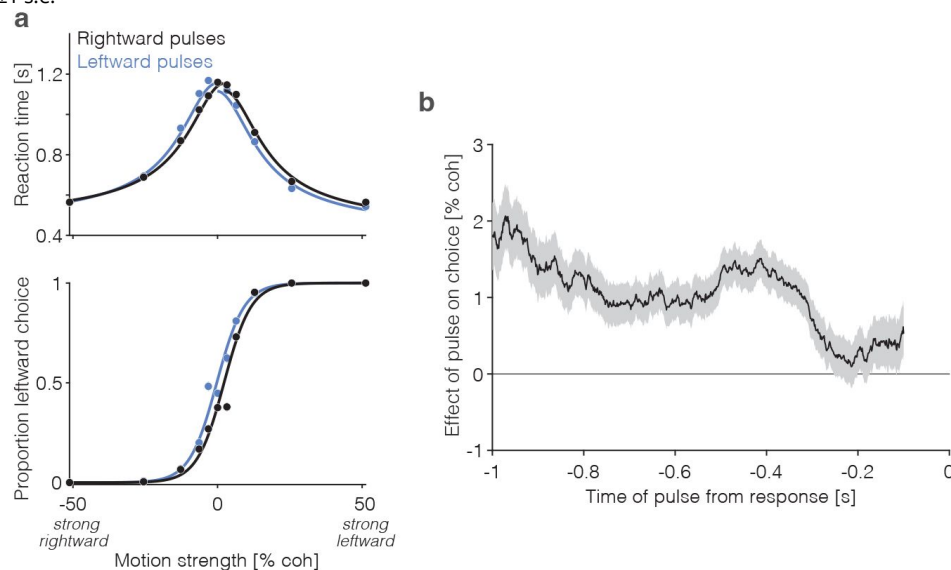
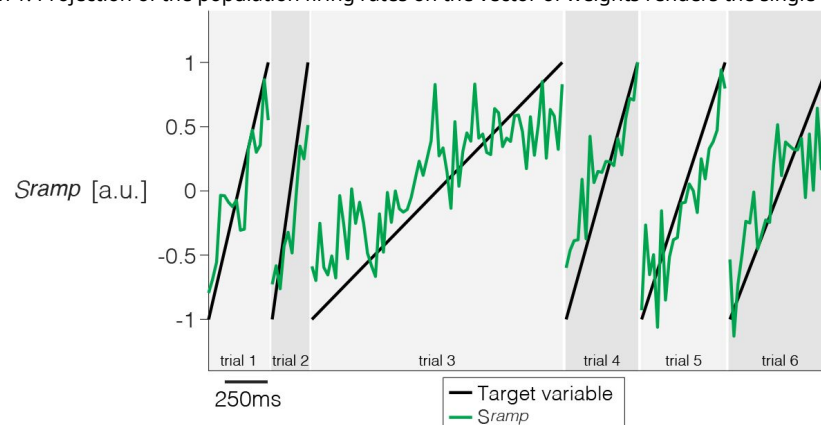


Figure S2.

Derivation of a ramp coding-direction in neuronal state space.

Weights are assigned to each of the N simultaneously recorded neurons in each session using simple least squares regression to approximate a ramp from -1 to $+1$ on the interval from 200 ms after motion onset to 100 ms before saccade initiation. Only trials ending in left (contraversive) choices are included. The graph shows the quality of the regression on 6 trials from Session 1. Projection of the population firing rates on the vector of weights renders the single trial signal, $S^{\text{ramp}}(t)$.



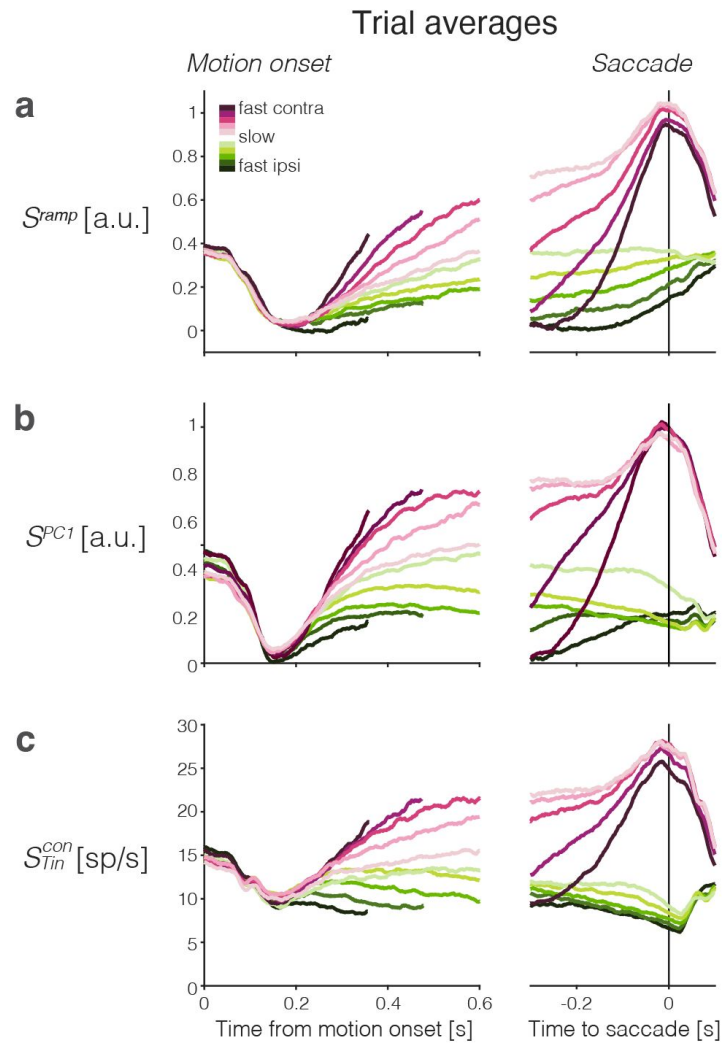


Figure S3.

Trial-averaged activity grouped by RT quantile.

Rows show the averages of single-trial responses of three signals. Choice and RT quantile are indicated by color (legend) and aligned to motion onset (left) and saccade initiation (right). **a**, Ramp coding direction. **b**, First principal component from PCA. **c**, Averages of the subset of neurons that have the left (contralateral) target in their response field. Correct and error trials are included in both motion and saccade aligned averages.

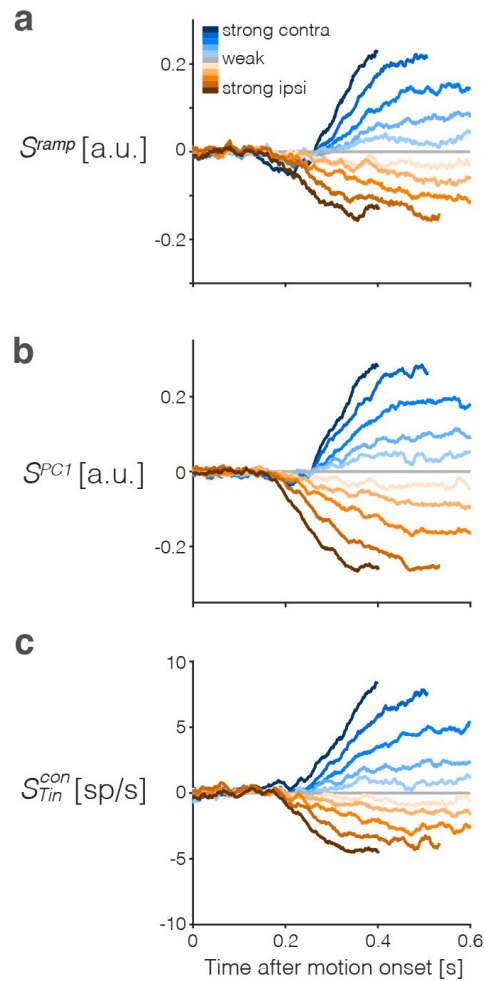


Figure S4.

Trial-averaged activity after subtracting the urgency component.

The urgency signal, $u(t)$, is a time-dependent, evidence-independent component of the neural activity that is thought to implement the equivalent of a collapsing bound in the race-model architecture of drift-diffusion shown in (Fig. 1c). We estimate $u(t)$ for each signal, S^x , as the average $S^x(t)$, aligned to motion onset, using only the 0% coherence motion trials (gray traces in the third column of Fig. 2). **a.** Ramp coding signal, $S^{\text{ramp}}(t)$, with urgency subtracted. **b.** PC1 signal, $S^{\text{PC1}}(t)$, with urgency subtracted. **c.** $S_{\text{Tin}}^{\text{con}}(t)$ with urgency subtracted.

Figure S5.

Bounds induce sublinear increase in variance of diffusion paths.

For unbounded diffusion (black) the variance across diffusion paths increases linearly with unity slope, and this holds under our smoothing procedure too. Symbols mark the median sample times of the first six non-overlapping $t \pm 25$ ms running means from the beginning of the epoch of integration, as in Fig. 3a. The red trace shows the values produced by stimulating the model illustrated in Fig. 1c (combined residuals from 20,000 trials per coherence -.032, 0, & +.032, as in Fig. 3a).

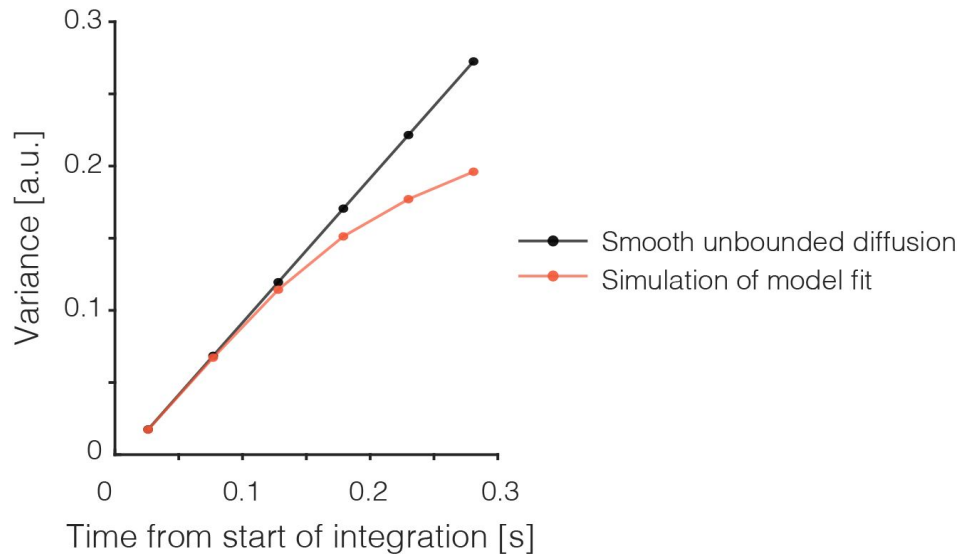
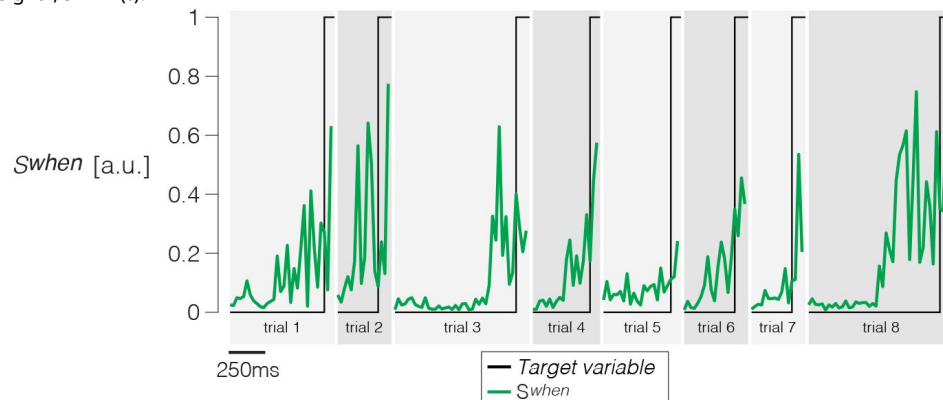


Figure S6.

Derivation of a *When* direction in neuronal state space.

Weights are assigned to each of the N simultaneously recorded neurons in each session using logistic regression to approximate a step that takes a value of 0 from 200 ms before motion onset to 150 ms before saccade initiation, and a value of 1 for the following 100 ms (the last 50 ms before the saccade are discarded). Only trials ending in left (contraversive) choices are included. The graph shows the quality of the regression on 8 example trials. The traces are shown from 200 ms after motion onset to 50 ms before the saccade. Projection of the population firing rates on the vector of weights renders the single trial signal, $S^{\text{When}}(t)$.



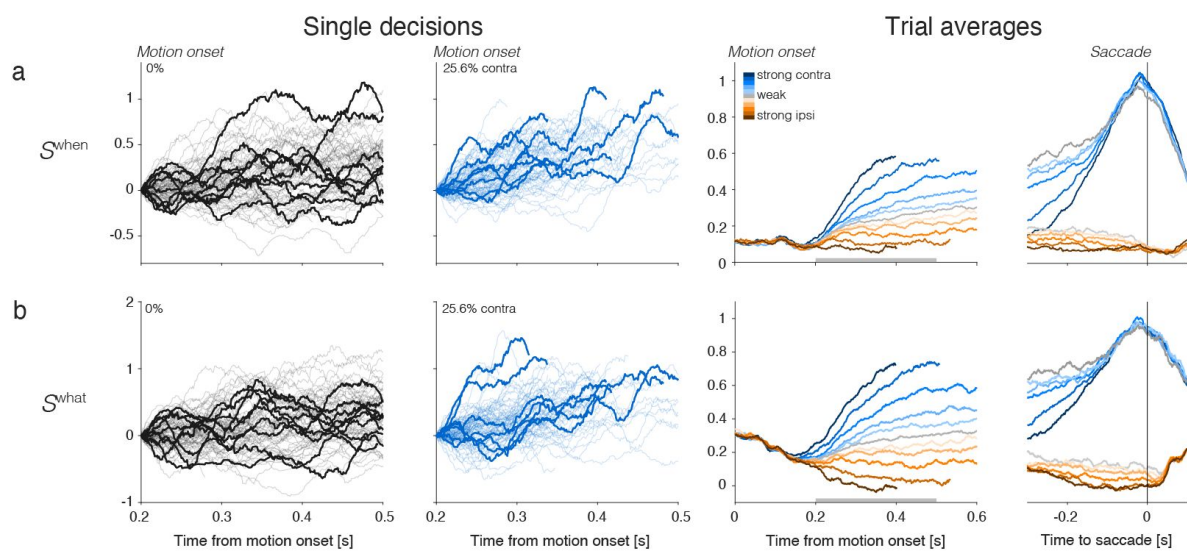


Figure S7.

Single-trials and trial averaged signals furnished by the When- and What-decoders.

Same convention as [Fig. 2](#), using the same single-trial examples.

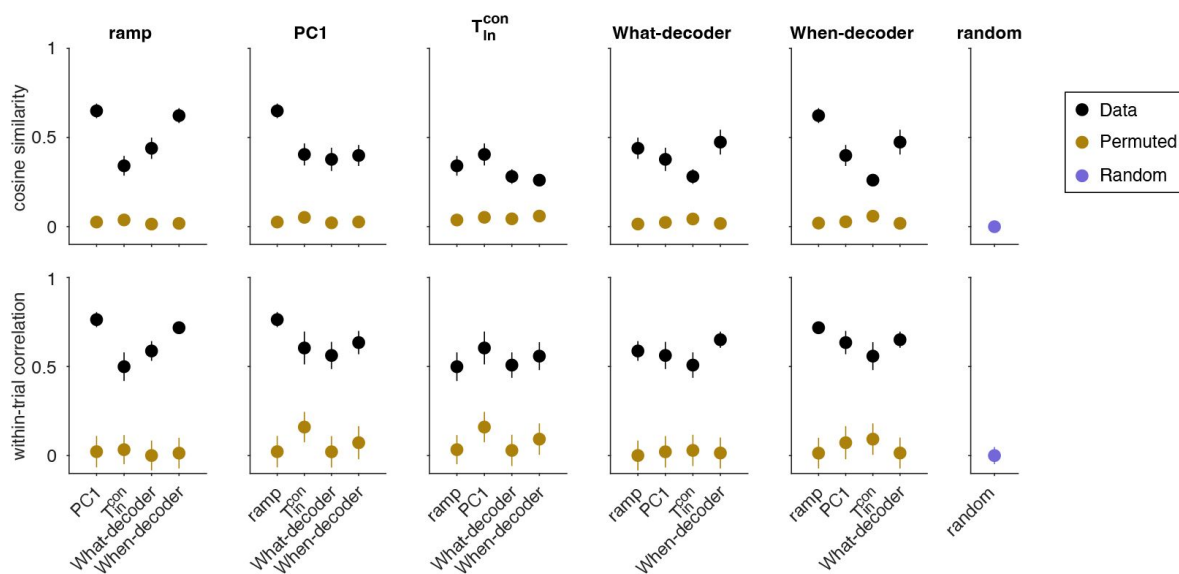


Figure S8.

Variability in cosine similarity and within-trial correlation across sessions.

Top. Cosine similarity (CS) of five coding directions. Black markers portray the same mean CS as in [Fig. 4d](#). Here, we also show the variability in each measure across sessions (s.e.m.). CS are significantly greater than zero between all pairs of CDs (all $p < 0.001$, t-test). Brown markers portray the same measure when the assignment between neurons and weights are permuted. Error bars reflect the standard deviation across 1000 such permutations. The purple marker in the rightmost column reflects the CS between random directions in state space. These vectors were generated by drawing from a Normal distribution, $\mathcal{N}(0, 1)$ and scaled to length 1. The CS in the data (black symbols) are significantly greater than the CS obtained from the two control analyses, for every pairwise combination of coding directions (permuted weights: all $p < 10^{-156}$, t-test; random unit vectors: all $p < 10^{-63}$). *Bottom.* Same conventions as *Top* for the within-trial correlations portrayed in [Fig. 4e](#). For each pair of coding directions, the within-trial correlations in the data are significantly greater than zero (all $p < 10^{-23}$). The correlations are also significantly greater than those between pairs of signals generated by projections of the data onto pairs of (i) random vectors established by permutations of the weights defining each coding direction (brown, all pairwise comparisons $p < 0.009$, z-test) and (ii) random unit vectors (purple, all pairwise comparisons $p < 10^{-14}$, z-test). This control serves mainly to refute the possibility that the correlations are explained by correlated variability in the neural population regardless of the signals produced by the weighted sums.

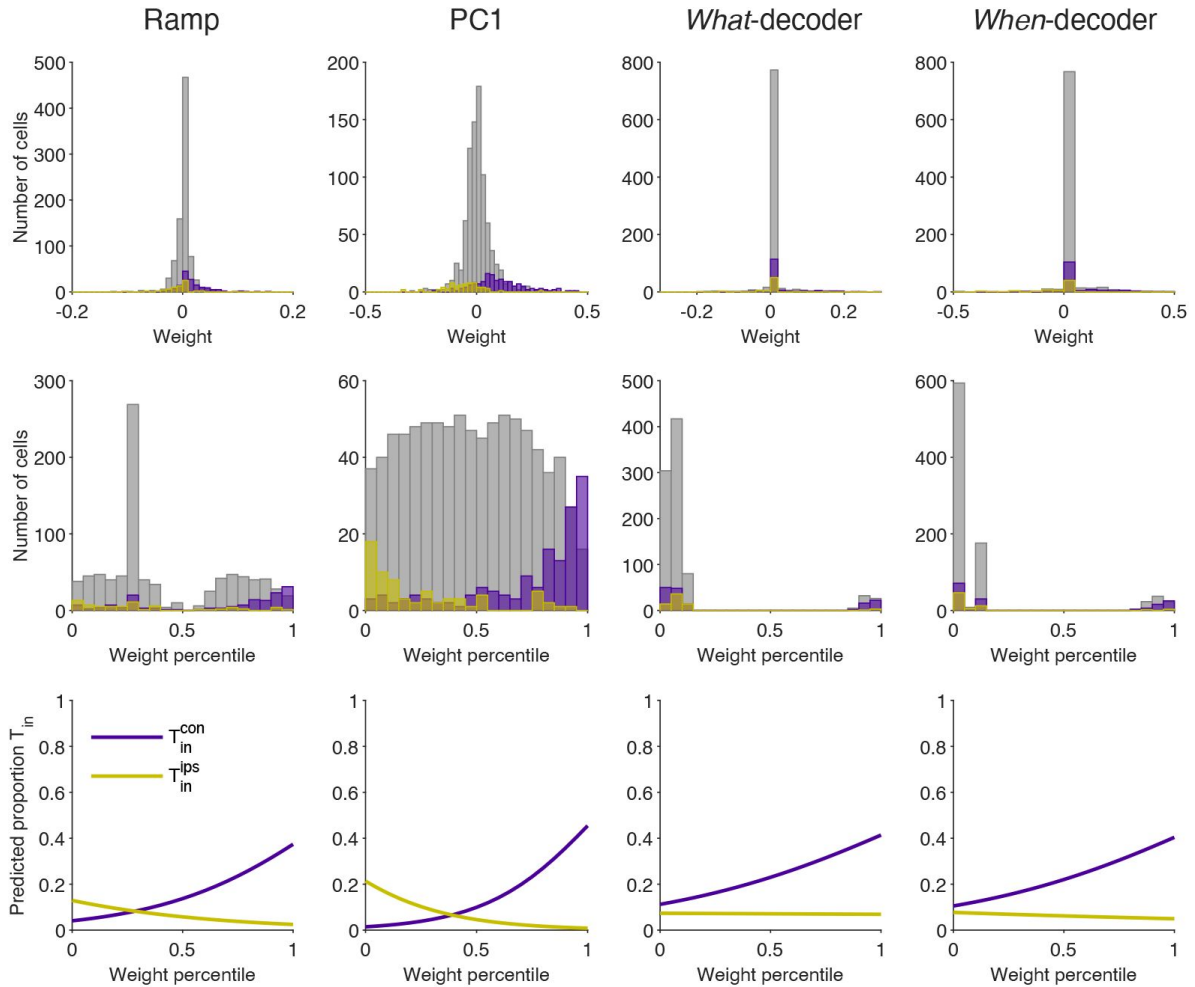


Figure S9.

The T_{in} neurons are not discoverable by their weight assignments.

The T_{in}^{con} neurons contribute strongly and positively to all coding directions, but they are a small fraction of the sampled population. Here we ask whether the rank of a neuron's weight might identify it as T_{in}^{con} . **a**, Distribution of weights assigned to T_{in}^{con} (purple), T_{in}^{ips} (yellow) and all other neurons (gray) for the coding directions specified by the column title. **b**, Same as **a** for weight percentiles (computed within session). **c**, The graphs are logistic fits $P(I_k) = \{1 + e^{-(\beta_0 + \beta_1 w_k)}\}^{-1}$, where $I = 1$ if neuron k is T_{in}^{con} , and $I_k = 0$ otherwise. Neurons with stronger positive or negative weights are more likely to be T_{in}^{con} or T_{in}^{ips} , respectively, but even the largest weight percentile identifies a T_{in}^{con} neuron with probability less than 0.4.

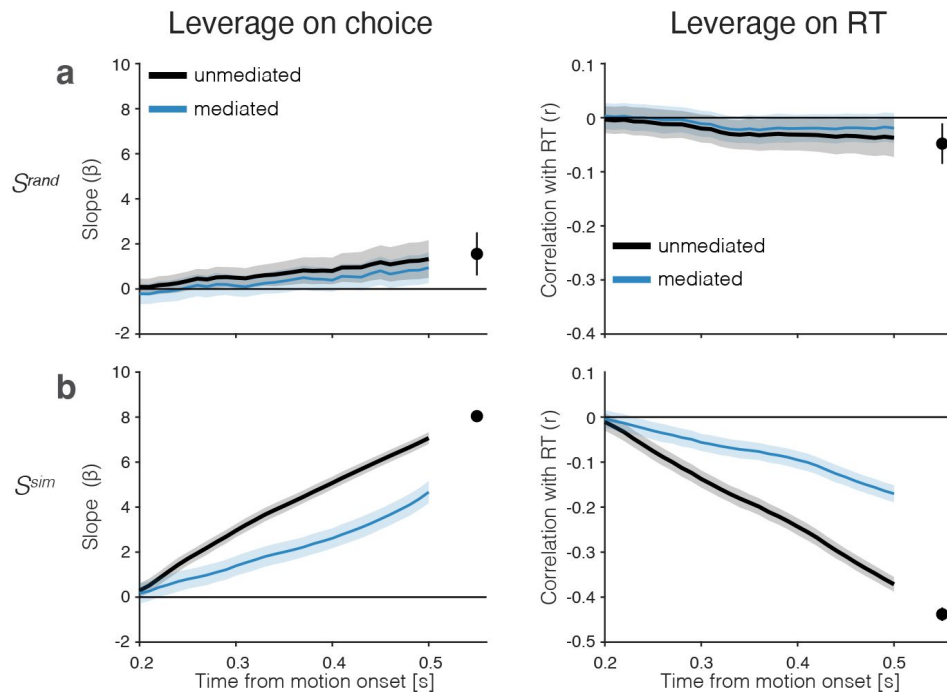


Figure S10.

Control analyses bearing on the leverage and mediation results in Fig. 5

a, Leverage of signals generated by projection of the neural data on random coding directions in neuronal state space (permutations of the PC1 weights). Same conventions and ordinate scale as in Fig. 5. **b**, Leverage of drift-diffusion signals derived from simulations of the racing drift-diffusion model fit to behavior (Fig. 1). The simulated signals establish the expectation of N weakly correlated noisy neurons. The traces agree qualitatively with the leverage and degree of mediation by T_{in}^{con} neurons. Importantly, these simulations highlight that the mediation of the leverage on behavior by a later sample is not complete unless the system is noise-free (for more details see Methods, *Leverage of single-trial activity on behavior*). Same conventions as in Fig. 5.

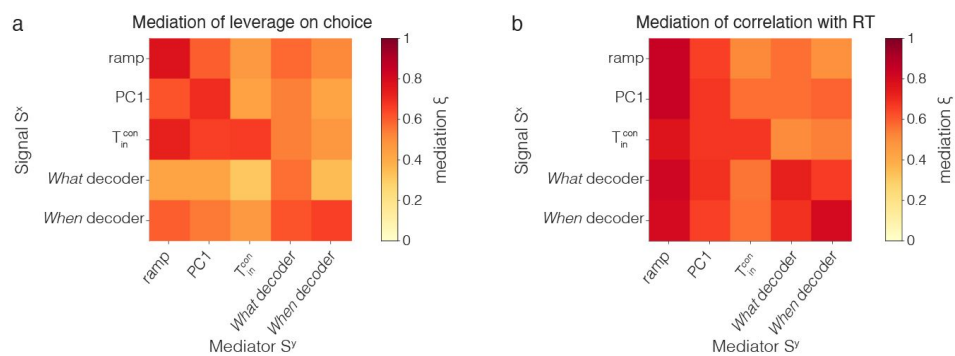


Figure S11.

Cross-mediation of Single-trial correlations with behavior.

The figure extends the observations in **Fig. 5** that a sample of S_{Tin}^{con} at $t = 550$ ms after motion onset (i) reduces the leverage of earlier samples of S_{Tin}^{con} on choice and RT (mediation of S^x on S^x) and (ii) also reduces the leverage of earlier samples of other signals, S^{PC1} and S^{ramp} (cross-mediation of S^x on S^y). The heatmaps are 5×5 matrices of the mediation indices, ξ^{Ch} and ξ^{RT} , (Eqs. 7 and 10), that is how a signal at $t = 550$ ms mediates a signal at $t = 400$ ms after motion onset. The index is zero if there is no mediation by the later sample; one of the mediation is complete. The main diagonal (top left to bottom right) shows the mediation of S^x (0.55) on S^x (0.4). Values below the diagonal show cross-mediation of S^y by S^x ; values above the diagonal show mediation of S^x by S^y . **a**, Leverage on choice. **b**, Leverage on RT. Notice that the matrices are not symmetric. For example, S^{ramp} mediates the leverage of S_{Tin}^{con} on choice more S_{Tin}^{con} mediates the leverage of S^{ramp} , and S_{Tin}^{con} mediates the leverage of S^{When} on RT more than S^{When} mediates the leverage of T_{in}^{con} .

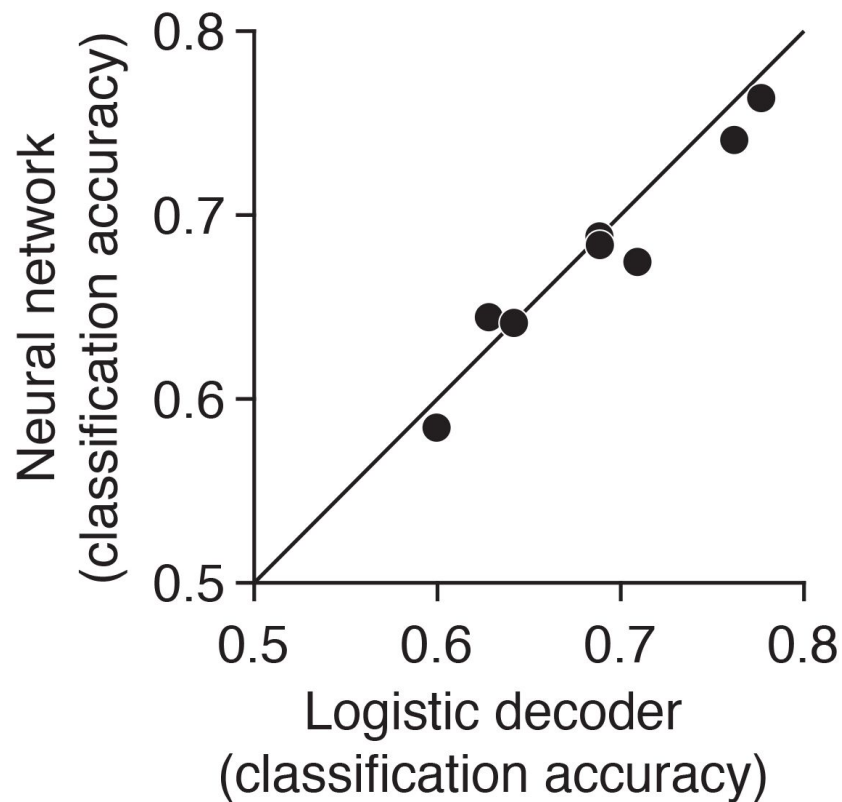


Figure S12.

Comparison of linear and non-linear choice decoders.

The figure depicts the cross-validated classification accuracy of two decoders that use population activity at $t = 450$ ms from motion onset to predict choice at the same time point on held out trials. The first is a linear decoder (logistic classifier, *abscissa*), as used for the *What* decoder in [Fig. 4](#). The second is a non-linear decoder (*ordinate*), which takes the form of a neural network with two hidden layers ($N_1 = 100$, $N_2 = 50$, units with sigmoid activation function). The two decoders perform similarly, with the Neural network outperforming the logistic decoder in only one of eight sessions. The analysis suggests that the assumption of linear embedding of the DV is justified.

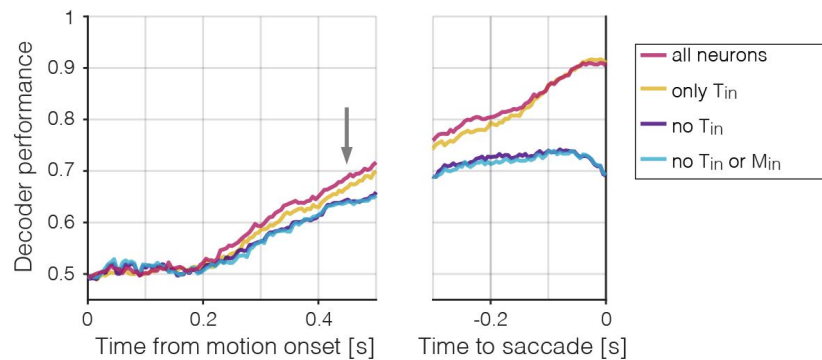


Figure S13.

Decoding choice from subsets of neurons.

Mean accuracy (across sessions) of four choice decoders, plotted as a function of time from motion onset (*left*) and time to saccadic choice (*right*). The decoders are trained on the neural activity between 425 and 475 ms from motion onset (gray arrow), and applied to all other time points (same method as **Fig. 4a**, *fixed training-time*). Colors indicate whether the decoders were trained using the activity of all neurons (red, same as in **Fig. 4a**), only T_{in}^{con} and T_{in}^{ips} neurons (yellow), all but T_{in} neurons (purple), or all but T_{in} and M_{in} neurons (blue). Decoding accuracy is diminished without the contribution of T_{in} neurons, which constitute only $21.7 \pm 2.1\%$ of the population (mean $\pm$ s.e.).

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Editors

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Michael Frank

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Reviewer #1 (Public Review):

Summary:

In this paper, Steinemann *et al.* characterized the nature of stochastic signals underlying the trial-averaged responses observed in lateral intraparietal cortex (LIP) of non-human primates (NHPs), while these performed the widely used random dot direction discrimination task. Ramp-up dynamics in the trial averaged LIP responses were reported in numerous papers before. But the temporal dynamics of these signals at the single-trial level have been subject to debate. Using large scale neuronal recordings with Neuropixels in NHPs, allows the authors to settle this debate rather compellingly. They show that drift-diffusion like computations account well for the observed dynamics in LIP.

Strengths:

This work uses innovative technical approaches (Neuropixel recordings in behaving macaque monkeys). The authors tackle a vexing question that requires measurements of

simultaneous neuronal population activity and hence leverage this advanced recording technique in a convincing way.

They use different population decoding strategies to help interpret the results.

They also compare how decoders relying on the data-driven approach using dimensionality reduction of the full neural population space compares to decoders relying on more traditional ways to categorize neurons that are based on hypotheses about their function. Intriguingly, although the functionally identified neurons are a modest fraction of the population, decoders that only rely on this fraction achieve comparable decoding performance to those relying on the full population. Moreover, decoding weights for the full population did not allow the authors to reliably identify the functionally identified subpopulation.

The revision addressed the minor weaknesses to our satisfaction.

<https://doi.org/10.7554/eLife.90859.2.sa2>

Reviewer #2 (Public Review):

Steinemann, Stine, and their co-authors studied the noisy accumulation of sensory evidence during perceptual decision-making using Neuropixels recordings in awake, behaving monkeys. Previous work has largely focused on describing the neural underpinnings through which sensory evidence accumulates to inform decisions, a process which on average resembles the systematic drift of a scalar decision variable toward an evidence threshold. The additional order of magnitude in recording throughput permitted by the methodology adopted in this work offers two opportunities to extend this understanding. First, larger-scale recordings allow for the study of relationships between the population activity state and behavior without averaging across trials. The authors' observation here of covariation between the trial-to-trial fluctuations of activity and behavior (choice, reaction time) constitutes interesting new evidence for the claim that neural populations in LIP encode the behaviorally-relevant internal decision variable. Second, using Neuropixels allows the authors to sample LIP neurons with more diverse response properties (e.g. spatial RF location, motion direction selectivity), making the important question of how decision-related computations are structured in LIP amenable to study. For these reasons, the dataset collected in this study is unique and potentially quite valuable. This revised manuscript addresses a number of questions regarding analyses which were unclear in the original manuscript, and as a result the study is a strong contribution toward our understanding of neural mechanisms of decision making.

<https://doi.org/10.7554/eLife.90859.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

In this paper, Steinemann et al. characterized the nature of stochastic signals underlying the trial-averaged responses observed in the lateral intraparietal cortex (LIP) of non-human primates (NHPs), while these performed the widely used random dot direction

discrimination task. Ramp-up dynamics in the trial averaged LIP responses were reported in numerous papers before. However, the temporal dynamics of these signals at the single-trial level have been subject to debate. Using large-scale neuronal recordings with Neuropixels in NHPs, allows the authors to settle this debate rather compellingly. They show that drift-diffusion-like computations account well for the observed dynamics in LIP.

Strengths:

This work uses innovative technical approaches (Neuropixel recordings in behaving macaque monkeys). The authors tackle a vexing question that requires measurements of simultaneous neuronal population activity and hence leverage this advanced recording technique in a convincing way

They use different population decoding strategies to help interpret the results.

They also compare how decoders relying on the data-driven approach using dimensionality reduction of the full neural population space compare to decoders relying on more traditional ways to categorize neurons that are based on hypotheses about their function. Intriguingly, although the functionally identified neurons are a modest fraction of the population, decoders that only rely on this fraction achieve comparable decoding performance to those relying on the full population. Moreover, decoding weights for the full population did not allow the authors to reliably identify the functionally identified subpopulation.

Weaknesses:

No major weaknesses beyond a few, largely clarification issues, detailed below.

We thank Reviewer 1 (R1) for this summary. The revised manuscript incorporates R1's suggestions, as detailed below.

Reviewer #2 (Public Review):

Steinemann, Stine, and their co-authors studied the noisy accumulation of sensory evidence during perceptual decision-making using Neuropixels recordings in awake, behaving monkeys. Previous work has largely focused on describing the neural underpinnings through which sensory evidence accumulates to inform decisions, a process which on average resembles the systematic drift of a scalar decision variable toward an evidence threshold. The additional order of magnitude in recording throughput permitted by the methodology adopted in this work offers two opportunities to extend this understanding. First, larger-scale recordings allow for the study of relationships between the population activity state and behavior without averaging across trials. The authors' observation here of covariation between the trial-to-trial fluctuations of activity and behavior (choice, reaction time) constitutes interesting new evidence for the claim that neural populations in LIP encode the behaviorally-relevant internal decision variable. Second, using Neuropixels allows the authors to sample LIP neurons with more diverse response properties (e.g. spatial RF location, motion direction selectivity), making the important question of how decision-related computations are structured in LIP amenable to study. For these reasons, the dataset collected in this study is unique and potentially quite valuable.

However, the analyses at present do not convincingly support two of the manuscript's key claims: (1) that "sophisticated analyses of the full neuronal state space" and "a simple average of Tconin neurons" yield roughly equivalent representations of the decision variable; and (2) that direction-selective units in LIP provide the samples of instantaneous evidence that these Tconin neurons integrate. Supporting claim (1) would require results

from sophisticated population analyses leveraging the full neuronal state space; however, the current analyses instead focus almost exclusively on 1D projections of the data. Supporting claim (2) convincingly would require larger samples of units overlapping the motion stimulus, as well as additional control analyses.

We thank the reviewer (R2) for their careful reading of our paper and the many useful suggestions.

As detailed below, the revised manuscript incorporates new control analyses, improved quantification, and statistical rigor, which now provide compelling support for key claim #1. We do not regard claim #2 as a key claim of the paper. It is an intriguing finding with solid support, worthy of dissemination and further investigation. We have clarified the writing on this matter.

Specific shortcomings are addressed in further detail below:

(1) The key analysis-correlation between trial-by-trial activity fluctuations and behavior, presented in Figure 5 is opaque, and would be more convincing with negative controls. To strengthen the claim that the relationship between fluctuations in (a projection of) activity and fluctuations in behavior is significant/meaningful, some evidence should be brought that this relationship is specific - e.g. do all projections of activity give rise to this relationship (or not), or what level of leverage is achieved with respect to choice/RT when the trial-by-trial correspondence with activity is broken by shuffling.

We do not understand why R2 finds the analysis opaque, but we are grateful for the lucid recommendations. The relationships between fluctuations in neural activity and behavior are indeed “specific” in the sense that R2 uses this term. In addition to the shuffle control, which destroys both relationships (Reviewer Figure 1), we performed additional control analyses that preserve the correspondence of neural signals and behavior on the same trial. We generated random coding directions (CDs) by establishing weight vectors that were either chosen from a standard normal distribution or by permuting the weights assigned to PC-1 in each session. The latter is the more conservative measure. Projections of the neural responses onto these random coding directions render $\text{Srand}(r)$. Specifically, the degree of leverage is effectively zero or greatly reduced. These analyses are summarized in a new Supplementary Figure S10. The bottom row of Figure S10 also addresses the question, “What degree of leverage and mediation would be expected for a theoretical decision variable?” This is accomplished by simulating decision variables using the drift-diffusion model fits in Figure 1c. The simulation is consistent with the leverage and (incomplete) mediation observed for the populations of Tcon neurons. For details see Methods, §§Simulated decision variables and Leverage in

of single-trial activity on behavior.

(2) The choice to perform most analysis on 1D projections of population activity is not wholly appropriate for this unique type of dataset, limiting the novelty of the findings, and the interpretation of similarity between results across choices of projection appears circular:

We disagree with the characterization of our argument as circular, but R2 raises several important points that will probably occur to other careful readers. We address them as subpoints 2.1–2.4, below. Importantly, we are neither claiming nor assuming that the LIP population activity is one-dimensional. We have revised the paper to avoid giving this impression. We are also not claiming that the average of Tin neurons (or the 1D projections) explains all features of the LIP population, nor would we expect it to, given the diversity of response fields across the population. Our objective is to identify the specific dimension

within population activity that captures the decision variable (DV), which has been characterized successfully as a one-dimensional stochastic process—that is, a scalar function of time. We have endeavored to clarify our thinking on this point in the revised manuscript (e.g., lines 97–98, 103–104).

(2.1) The bulk of the analyses (Figure 2, Figure 3, part of Figure 4, Figure 5, Figure 6) operate on one of several 1D projections of simultaneously recorded activity. Unless the embedding dimension of these datasets really does not exceed 1 (dimensionality using e.g. participation ratio in each session is not quantified), it is likely that these projections elide meaningful features of LIP population activity.

We now report the participation ratio (4.4 ± 0.4 , mean $\pm$ s.e. across sessions), and we state that the first 3 PCs explain $67.1 \pm 3.1\%$ of the variance of time- and coherence-dependent signals used for the PCA. We agree that the 1D projections may elide meaningful features of LIP population activity. Indeed, we make this point through our analysis of the Min neurons. We do not claim that the 1D projections explain all of the meaningful features of LIP population activity. They do, however, reveal the decision variable, which is our main focus. These 1D signals contain features that correlate with events in the superior colliculus, summarized in Stine et al. (2023), attesting to their biological relevance.

(2.2) Further, the observed similarity of results across these 1D projections may not be meaningful/interpretable. First, the rationale behind deriving Sramp was based on the ramping historically observed in Tin neurons during this task, so should be expected to resemble Tin.

The Reviewer is correct that we would expect Sramp to resemble the ramping observed in Tin neurons. We refer to this approach as hypothesis-driven. It captures the drift component of drift-diffusion. It is true that the Tcon neurons exhibit such ramps in their trial average firing rates, but this does not guarantee in

that the single-trial population firing rates would manifest as drift-diffusion. Indeed Latimer et al. (2015) concluded that the ramp-like averages comprise stepping from a low to a high firing rate on each trial at a random time. Therefore, while R2 is right to characterize the similarity of Tcon to the ramp direction in in trial-averaged activity as unsurprising, their similarity on single trials is not guaranteed.

(2.3) Second, Tin comprises the largest fraction of the neuron groups sampled during most sessions, so SPC1 should resemble Tin too. The finding that decision variables derived from the whole population's activity reduce essentially to the average of Tin neurons is thus at least in part 'baked in' to the approach used for deriving the decision variables.

This is incorrect. The Tcon in neurons constitute only 14.5% of the population, on average, across the sessions (see Table 1). This misunderstanding might contribute to R2's concern about the importance of these neurons in shaping PC1. It is not simply because they are over-represented. Also, addressing R2's concern about circularity, we would like to remind R2 that the selection of Tin neurons was based only on their spatial selectivity in the delayed saccade task. We do not see how it could be baked-in/guaranteed that a simple average of these neurons (i.e. zero degrees of freedom) yields dynamics and behavioral correlations that match those produced by dimensionality-reduction techniques that (i) have degrees of freedom equal to the number of neurons and (ii) are blind to the neurons' spatial selectivity. We have additionally modified what is now Supplementary Figure S13 (old Supplementary Figure S8), which portrays the mean accuracy of choice decoders trained on the neural activity of all neurons, only Tin neurons, all but the Tin neurons, and all but Tin and Min

neurons, respectively. Figure S13 now highlights how much more readily choice can be decoded from the small population of Tin neurons than the remainder of the population.

(2.4) The analysis presented in Figure S6 looks like an attempt to demonstrate that this isn't the case, but is opaque. Are the magnitudes of weights assigned to units in Tin larger than in the other groups of units with preselected response properties? What is their mean weighting magnitude, in comparison with the mean weight magnitude assigned to other groups? What is the null level of correspondence observed between weight magnitude and assignment to Tin (e.g. a negative control, where the identities of units are scrambled)?

The revised Figure S6—what is now Figure S9—displays more clearly that the weights assigned to Tcon and Tips neurons (purple & yellow, respectively) are larger in magnitude than those assigned in in to other neurons (gray). Author response table 1 shows a more detailed breakdown of the groups. Note that the length of the vector of weights is one. We are unsure what R2 means by “the null level of correspondence.” Perhaps it helps to know that the mean weight of the “other neurons” is close to zero for all four coding directions. However, it is the overlap of the weights and the relative abundance of non-Tin neurons that is more germane to the point we are making. To wit, knowing the weight (or percentile) of a neuron is a poor predictor that it belongs to the Tin category. This point is most clearly supported by the logistic regression (Fig. S9, bottom row). In other words, the large group of non-Tin neurons contribute substantially to all four coding directions examined in Figure S9. Thus, the similarity between Tin neurons and PC1 is not simply due to an over-representation of Tin neurons as suggested in item 2.3.

Author response table 1.

Mean weights assigned to neuron classes in four coding directions.

Neuron class	Sramp	SPC1	SWhen	SWhat
wTinC	0.073	0.093	0.063	0.059
wTinI	-0.035	-0.063	-0.027	-0.022
wMinL	0.011	0.012	0.028	0.027
wMinR	-0.035	-0.043	-0.027	0.014
wOther	0.0010	0.0051	0.0020	0.0070

(3) The principal components analysis normalization procedure is unclear, and potentially incorrect and misleading: Why use the chosen normalization window (± 25 ms around 100ms after motion stimulus onset) for standardizing activity for PCA, rather than the typical choice of mean/standard deviation of activity in the full data window? This choice would specifically squash responses for units with a strong visual response, which distorts the covariance matrix, and thus the principal components that result. This kind of departure from the standard procedure should be clearly justified: what do the principal components look like when a standard procedure is used, and why was this insufficient/incorrect/unsuitable for this setting?

We used the early window because it is a robust measure of overall excitability, but we now use a more conventional window that spans the main epoch of our analyses, 200–600 ms

after motion onset. This method yields results qualitatively similar to the original method. We are persuaded that this is the more sensible choice. We thank R2 for raising this concern.

(4) Analysis conclusions would generally be stronger with estimates of variability and control analyses: This applies broadly to Figures 2-6.

We have added estimates of variability and control analyses where appropriate.

Figure 2 shows examples of single-trial signals. The variability is addressed in Figure 3a and the new Supplementary Figure S5.

Figure 3 now contains error bars derived by bootstrapping (see Methods, §Variance and autocorrelation of smoothed diffusion signals). We have also added Supplementary Figure S5, which substantiates the sublinearity claim using simulations.

Figure 4 (i) We now indicate the s.e.m. of decoding accuracy (across sessions) by the shading in Figure 4a. (ii) The black symbols in new Supplementary Figure S8 show the mean $\pm$ s.e.m. for all pairwise comparisons shown in Figure 4d & e. (iii) Supplementary Figure S8 also summarizes two control analyses that deploy random coding directions (CDs) in neuronal state space. The upper row of Fig S9 compares the observed cosine similarity (CoSim)—between the CD identified by the graph title and the other four CDs labeled along the abscissa—with values obtained with 1000 random CDs established by random permutations of the weight assignments. The brown symbols are the mean $\pm$ sdev of the CoSim (N=1000). The error bars are smaller than the symbols. We use the cumulative distribution of CoSim under permutation to estimate p-values ($p < 0.001$ for all comparisons). We used a similar approach to estimate the distribution of the analogous correlation statistics between signals rendered by random directions in state space (Figure S8, lower row). For additional details, please see Methods, §Similarity of single-trial signals.

Figure 5: The rigor of all claims associated with this figure is added from two control analyses and a simulation. The first control breaks the trial-by-trial correspondence between neural signals and behavior (Reviewer Figure 1). The second control shows that neural activity does not have substantial leverage on behavior when projected onto random directions in state space (Supplementary Figure S10, top). Simulations of decision variables using parameters derived from the fits to the behavioral data (Figure 1) support a degree of leverage and mediation comparable to the values observed for $\mathcal{S}$ Tincon (Supplementary Figure S10, bottom). For additional details, please see Methods (§Leverage of single-trial activity on behavior) and the reply to item 1, above.

Figure 6: Panels c&d show estimates of variability across neurons and experimental sessions, respectively. The reported p-value is based on a permutation test (see Methods, §Correlations between Min and Tconin). The correlations shown in panel e (heatmap) are derived from pooled data across sessions. The reported p-value is based on a permutation test (see Methods, §Correlations between Min and Tconin).

Reviewer #3 (Public Review):

Summary:

The paper investigates which aspects of neural activity in LIP of the macaque give rise to individual decisions

(specificity of choice and reaction times) in single trials, by recording simultaneously from hundreds of neurons. Using a variety of dimensionality reduction and decoding techniques, they demonstrate that a population-based drift-diffusion signal, which relies on a small subset of neurons that overlap choice targets, is responsible for the choice and reaction time variability. Analysis of direction-selective neurons in LIP and their

correlation with decision-related neurons (T_{con} in $[T_{conin}]$ neurons) suggests that evidence integration occurs within area LIP.

Strengths:

This is an important and interesting paper, which resolves conflicting hypotheses regarding the mechanisms that underlie decision-making in single trials. This is made possible by exploiting novel technology (Primatepixels recordings), in conjunction with state-of-the-art analyses and well-established dynamic random dot motion discrimination tasks.

General recommendations

(1) Please tone down causal language. You present compelling correlative evidence for the idea that LIP population activity encodes the drift-diffusion DV. We feel that claims beyond that (e.g., "Single-trial drift-diffusion signals control the choice and decision time") would require direct interventions, and are only partially supported by the current evidence. Further examples are provided in point 1) of Reviewer 1 below.

We have adopted the recommendation to "tone down the causal language." Throughout the manuscript, we strive to avoid conveying the false impression that the present findings provide causal support for the decision mechanism. However, other causal studies of LIP support causality in the random dot motion task (Hanks et al., 2006; Jeurissen et al., 2022). It is therefore justifiable to use terms that imply causality in statements intended to convey hypotheses about mechanism. We agree that we should not give the false impression that the present support for said mechanism is adduced from causal perturbations in this study, as there were none.

(2) Please provide a commonly used, data-driven quantification of the dimensionality of the population activity – for example, using participation ratio or the number of PCs explaining 90 % of the variance. This will help readers evaluate the conclusions about the dimensionality of the data.

Principal component analysis reveals a participation ratio of 4.4 ± 0.4 (mean $\pm$ s.e., across sessions), and the first 3 PCs explain 67.1 ± 3.1 percent of the variance. The dimensionality of the data is low, but greater than one. We state this in Methods (§Principal Component Analysis) and in Results (§Single-trial drift-diffusion signals approximate the decision variable, lines 200–201).

(3) Please justify the normalization procedure used for PCA: Why use the chosen normalization window (± 25 ms around 100 ms after motion stimulus onset) for standardizing activity for PCA, rather than the more common quantification of mean/standard deviation across the full data window? What do the first principal components look like when the latter procedure is used?

We now use a more conventional window that spans the main epoch of our analyses, 200–600 ms after motion onset. This method yields results qualitatively similar to the original method. We are persuaded that this is the more sensible choice.

(4) Please provide estimates of variability for variance and autocorrelation in Fig. 3 (e.g., through bootstrapping). Further, simulations could substantiate the claim about the expected sub-linearity at later time points (Fig. 3a) due to the upper stopping bound and limited firing rate range.

We thank the reviewers for these helpful recommendations. The revised Fig. 3 now contains error bars derived by bootstrapping (see Methods, §Variance and autocorrelation of smoothed diffusion signals). We have also added Supplementary Figure S5, which substantiates the sub-linearity claim using simulations.

(5) Please add controls and estimates of variability for decoding across sessions in Fig. 4: what are the levels of within-trial correlation/cosine similarity for random coding directions? What is the variability in the estimates of values shown in a/d/e?

We have addressed each of these items. (1) Figure 4a now shows the s.e.m. of decoding accuracy (across sessions). (2) Regarding the variability of estimates shown in Figure 4d & e, the standard errors are displayed in the new supplementary Figure S8. It makes sense to show them there because there is no natural way to represent error on the heat maps in Figure 4, and Figure S8 concerns the comparison of the values in Figure 4d&e to values derived from random coding directions. (3) Random coding directions lead to values of cosine similarity and within-trial correlation that do not differ significantly from zero. We show this in several ways, summarized in our reply to Public Review item 4. Additional details are in the revised manuscript (Methods, §Similarity of single-trial signals) and the new Supplementary Figure S8.

(6) Please perform additional analysis to strengthen the claim from Fig. 6, that Min represents the integrand and not the integral. The analysis in Fig. 6d could be repeated with the integral (cumulative sum) of the single-trial Min signals. Does this yield an increase in leverage over time?

The short answer is, yes in part. Reviewer Figure 2a provides support for leverage of the integral on choice, and this leverage, like $STin(t)$, increases as a function of time. The effect is present in all seven sessions that have both Mleftin and Mrightin neurons (all $p < 1e - 10$). However, as shown in panel b, the same integral fails to demonstrate more than a hint of leverage on RT. All correlations are barely negative, and the magnitude does not increase as a function of time. We suspect—but cannot prove—that this failure arises because of limited power and the expected weak effect. Recall that the mediation analysis of RT is restricted to longer trials. Moreover, the correlation between the Min difference and the Tin signal is less than 0.1 (heatmap, Fig. 6e), implying that the Min difference explains less than 1% of the variance of $STin(t)$. We considered including Reviewer Figure 2 in the paper, but we feel it would be disingenuous (cherry-picking) to report only the positive outcome of the leverage on choice. If the editors feel strongly about it, we would be open to including it, but leaving these analyses out of the revised manuscript seems more consistent with our effort to deemphasize this finding. In the future, we plan to record simultaneously from populations MT and LIP neurons (Min and Tin, of course) and optimize Min neuron yield by placing the RDM stimulus in the periphery.

(7) Please describe the complete procedure for determining spatially-selective activity. E.g.: What response epoch was used, what was the spatial layout of the response targets, were responses to all ipsi- vs contralateral targets pooled, what was the spatial distribution of response fields relative to the choice targets across the population?

We thank the reviewers for pointing out this oversight. We now explain this procedure in the Methods (lines 629–644):

Neurons were classified post hoc as Tin by visual-inspection of spatial heatmaps of neural activity acquired in the delayed saccade task. We inspected activity in the visual, delay, and perisaccadic epochs of the task. The distribution of target locations was guided by the spatial selectivity of simultaneously recorded neurons in the superior colliculus (see Stine 2023 for

details). Briefly, after identifying the location of the SC response fields, we randomly presented saccade targets within this location and seven other, equally spaced locations at the same eccentricity. In monkey J we also included 1–3 additional eccentricities, spanning 5–16 degrees. Neurons were classified as Tin if they displayed a clear, spatially-selective response in at least one epoch to one of the two locations occupied by the choice targets in the main task. Neurons that switched their spatial selectivity in different epochs were not classified as Tin. The classification was conducted before the analyses of activity in the motion discrimination task. The procedure was meant to mimic those used in earlier single-neuron studies of LIP (e.g., Roitman & Shadlen 2002) in which the location of the choice targets was

determined online by the qualitative spatial selectivity of the neuron under study. The Tcon neurons in the in present study were highly selective for either the contralateral or ipsilateral choice target used in the RDM task ($AUC = 0.89 \pm 0.01$; $p < 0.05$ for 97% of neurons, Wilcoxon rank sum test). Given the sparse sampling of saccade target locations, we are unable to supply a quantitative estimate of the center and spatial extent of the RFs.

(8) Please clarify if a neuron could be classified as both Tin and Min. Or were these categories mutually exclusive?

These categories are mutually exclusive. If a neuron has spatially-selective persistent activity, as defined by the method described above, it is classified as a Tin neuron and not as a Min neuron even if it also shows motion-selective activity during passive motion viewing. We now specify this in the Methods (lines 831–832).

Reviewer * 1, private recommendations

*R*1.1a Causal language (Line 23-24): “population activity represents [...] drift” and “we provide direct support for the hypothesis that drift-diffusion signal is the quantity responsible for the variability in choice and RT” reads at first sight as if the authors claim that they present evidence for a causal effect of LIP activity on choice. The authors are otherwise nuanced and careful to point out that their evidence is correlational. What seems to be meant is that the population activity/drift-diffusion signal “approximates the DV that gives rise to the choices [...]” (cf. line 399). I would recommend using such alternative phrasing to avoid confusion (and the typically strong reactions by readers against misleading causal statements).*

We have adopted the reviewer’s recommendation and have modified the text throughout to reduce causal language. See our response to General Recommendation 1.

*R*1.1b Relatedly, any discussion about the possibility of LIP being causally involved in evidence integration (e.g. lines 429-445 [Au: now 462–478]) should also comment on the possibility of a distributed representation of the decision variable given that neural correlates of the DV have been reported in several areas including PFC, caudate and FEF.*

We believe this is possible. However, we hope to avoid discussions about causality given that it is not a focus of the paper. Although it is somewhat tangential, we have shown elsewhere that LIP is causal in the sense that causal manipulations affect behavior, but it is also true that causality does not imply necessity, and similarly, lack of necessity does not imply “only correlation.” Regarding distributed representations, it is worth keeping in mind the cautionary counter-example furnished by the SC study (Stine et al., 2023). The firing rates measured by averaging over trials are similar in SC and LIP; both manifest as coherence and direction-dependent ramps, leading to the suggestion that they form a distributed representation of the decision variable. With single-trial resolution, we now know that LIP and SC exhibit distinct dynamics—drift-diffusion and bursting, respectively. It remains to be

seen if single-trial resolution achievable by simultaneous Neuropixels recordings from prefrontal areas and LIP reveal shared or distinct dynamics.

*R*1.2 How was the spatially selective activity determined? The classification of Tin neurons is critical to this study - how was their spatial selectivity determined? Please describe this in similar detail as the description of direction selectivity on lines 681-690 [Au: now 824-832]. E.g.: what response epoch was used, what was the spatial layout of the response targets, were responses to all ipsi- vs contralateral targets pooled, and what was the spatial distribution of response fields relative to the choice targets across the population?*

We now explain the selection procedure in Methods (lines 629-644). Please see our reply to General Recommendation 7, above.

*R*1.3 Could a neuron be classified as both Tin and Min, or were these categories mutually exclusive? Please clarify. (This goes beyond the scope of the current study: but did the authors find evidence for topographic organization or clustering of these categories of neurons?)*

These categories are mutually exclusive. Please see our response to General Recommendation 8, above.

*R*1.4 Contrary to the statement on line 121, the trial averages in Fig. 2a, 2b show coherence dependency at the time of the saccade in saccade-aligned traces for the coding strategies, except for STin (fig. 2c). Is this a result of the choice for t_1 (= 0.1s)? (The authors may want to change their statement on line 121.) Relatedly, do the population responses for the two coding strategies Sramp and SPC1 depend on the epoch used to derive weights for individual neurons?*

We have revised the description to accommodate R2's observation. Sramp retains weak coherence-dependence before saccades towards the choice target contralateral to the recording site. This was true in four of the eight sessions. For SPC1, there is no longer a coherence dependency for the Tin choices, owing to the change in normalization method (see revised Figure 2b).

We also corrected an error in the Methods section. Specifically, the ramp ends at $t_1 = 0.05$ s before the time of the saccade, not $t_1 = 0.1$ s. While we no longer emphasize the similarity of traces aligned to saccade, it is reasonable to find issue with the observation that they retain a dependency on coherence (Sramp only) because, according to theory, traces associated with Tin choices should reach a common positive threshold at decision termination. That said, for the Ramp direction there may be a reason to expect this discrepancy from theory. The deterministic part of drift-diffusion includes an urgency signal that confers positive convexity to the deterministic drift. This accelerating nonlinearity is not captured by the ramp, and it is more prominent at longer decision times, thus low coherences. We do not share this interpretation in the revised manuscript, in part because retention of coherence dependency is present in only half the sessions (see Reviewer Figure

(3) The correction to the definition of t_1 also provides an opportunity to address R2's final question ("Relatedly,...?"). For Sramp this particular variation in t_1 does not affect Sramp, and SPC1 no longer retains coherence dependency for Tin choices. Note that our choice of t_0 and t_1 is based on the empirical observation that the ramping activity in response averages of Tin neurons typically begins 200 ms after motion onset and ends 50-100 ms before initiation of the saccadic choice. The starting time (t_0) is also supported by the observation that the decoding accuracy of a choice-decoder begins to diverge from chance at this time (Figure 4a).

*R*1.5 It is intriguing that Sramp and SPC1 show dynamics that look so similar (fig. 2a, 2b). How do the weights assigned to each neuron in both strategies compare across the population?*

The weights assigned to each neuron are very similar across the two strategies as indicated by a cosine similarity (0.65 ± 0.04 , mean $\pm$ s.e.m. across sessions).

*R*1.6 Tin neurons, which show dynamics closely resembling different coding directions (fig. 2) and the decoders do not have weights that can distinguish them from the rest of the population in each of these analyses (fig. S7). Is it fair to interpret these findings as evidence for broad decision-related co-variability in the recorded neural population in LIP?*

Yes, our results are consistent with this interpretation. However, it is worth reiterating that decoding performance drops considerably when Tin neurons are not included (see Supplementary Figure S13). Thus, this broad decision-related co-variability is present but weak.

*R*1.7 It is intriguing that the decoding weights of the different decoders did not allow the authors to reliably identify Tin neurons. Could this be, in part, due to the low dimensionality of the population activity and task that the animals are presumably overtrained on? Or do the authors expect this finding to hold up if the population activity and task were higher dimensional?*

Great question! We can only speculate, but it seems possible that a more complex, “higher dimensional” task could make it easier to identify Tin neurons. For example, a task with four choices instead of two may decrease correlations among groups of neurons with different response fields. We have added this caveat to the discussion (lines 459–461). One minor semantic objection: The animal has learned to perform a highly contrived task at low signal-to-noise. The animal is well-trained, not over-trained.

*R*1.8 Lines 135-137 [Au: now 141–142]: The similarity in the single trial traces from different coding strategies (fig. 2a-2c, left) is not as evident to me as the authors suggest. It might be worthwhile computing the correlation coefficients between individual traces for each pair of strategies and reporting the mean correlation to support the author's point.*

We report the mean correlation between single-trial signals generated by the chosen dimensionality reduction methods in Figure 4e. We show the variability in this measure in Supplementary Figure S8. We have also adjusted the opacity of the single-trial traces in Figure 2, left.

*R*1.9 Minor/typos:*

-line 74: consider additionally citing Hyafil et al. 2023.

-line 588: “that were strongly correlated”?

-line 615: “were the actual drift-diffusion process were...”.

-line 717: “a causal influence” -> “no causal influence”.

Fig. 6: panel labels e vs d are swapped between the figure and caption.

Fig. 3c: labels r1,3 & r2,3 are flipped.

We have addressed all of these items. Thank you.

Reviewer 2 ✖, private recommendations

*R*2.1 (Figure 2) Determine whether restricting the analysis to 1D projections of the data is a suitable approach given the actual dimensionality of the datasets being analyzed:*

- Should show some quantification of the dimensionality of the recorded activity; could do this by quantifying the dimensionality of population activity in each session, e.g. with participation ratio or related measures (like # PCs to explain some high proportion of the variance, e.g. 90 %). If much of the variation is not described in 1 dimension, then the paper would benefit from some discussion/analysis of the signals that occupy the other dimensions.

We now report the participation ratio (4.4 ± 0.4 , mean $\pm$ s.e. across sessions), and we state that the first 3 PCs explain $67.1 \pm 3.1\%$ of the variance of the time- and coherence-dependent signals used for the PCA (mean $\pm$ s.e.). We agree that the 1D projections may elide meaningful features of LIP population activity. Indeed, we make this point through our analysis of the Min neurons. To reiterate our response above, we do not claim that the 1D projections explain all of the meaningful features of LIP population activity. They do, however, reveal the decision variable, which is our main focus. These 1D signals contain features that correlate with events in the superior colliculus, summarized in Stine et al. (2023), attesting to their biological relevance.

The Reviewer is correct that our approach presupposes a linear embedding of the 1D decision variable in the population activity. In other words, a non-linear representation of the 1D decision variable in population activity could have an embedding dimensionality greater than 1, and there may well be a non-linear method that reveals this representation. To test this possibility, we decoded choice on each trial from population activity using (1) a linear decoder (logistic classifier) or (2) a multi-layer neural network, which can exploit non-linearities. We found that, for each session, the two decoders performed similarly: the neural network outperforms the logistic decoder (barely) in just one session. The analysis suggests that the assumption of linear embedding of the decision variable is justified. We hope this analysis convinces the reviewer that “sophisticated analyses of the full neuronal state space” and “a simple average of [Tcon] neurons” do in indeed yield roughly equivalent representations of the decision variable. We have included the results of this analysis in Supplementary Figure S12. See also item 2 of the Public response.

*R*2.2 (Figure 3) Add estimates of variability for variance and autocorrelation through time from single-trial signals:*

- E.g. by bootstrapping. Would be helpful for making rigorous the discussion of when the deviation from the theory is outside what would be expected by chance, even if it doesn't change the specific conclusions here.

- If possible, it would help (by simulations, or maybe an added reference if it exists) to substantiate the claim about the expected sub-linearity at later time-points (Figure 3a) due to the upper stopping bound and limited firing rate range.

We thank the reviewer for this helpful comment. The revised Fig. 3 now contains error bars derived by bootstrapping (see Methods, §Variance and autocorrelation of smoothed diffusion signals). We have also added Supplementary Figure S5, which substantiates the sub-linearity claim using simulations.

*R*2.3 (Figure 4) Add controls and estimates of variability for decoding across sessions:*

- As a baseline - what is the level of within-trial correlation/cosine similarity when random coding directions are used?*
- What is the variability in the estimates of values shown in a/d/e?*

We have addressed each of these items. (1) Figure 4a now shows the s.e.m. of decoding accuracy (across sessions). (2) Regarding the variability of estimates shown in Figure 4d & e, the standard errors are displayed in the new Supplementary Figure S8. It makes sense to show them there because (i) there is no natural way to represent error on the heat maps in Figure 4, and (ii) S8 concerns the comparison of the values in Figure 4d & e to values derived from random coding directions. (3) Random coding directions lead to values of cosine similarity and within-trial correlation that do not differ significantly from zero. We show this in several ways, summarized in our reply to Public Review item 4. Additional details are in the revised manuscript (Methods: Similarity of single-trial signals) and the new Supplementary Figure S8. We also provide this information in response to Recommendation 5, above.

*R*2.4 (Figure 5) Add negative controls and significance tests to support claims about trends in leverage:*

- What is the level of increase in leverage attained from random 1D projections of the data, or other projections where the prior would be no leverage?*
- What is the range of leverage values fit for a simulated signal with a ground-truth of no trend?*

We have added two control analyses. In addition to a shuffle control, which destroys the relationship (Review Figure 1) we performed additional analyses that preserve the correspondence of neural signals and behavior on the same trial. We generated random coding directions (CDs) by establishing weight-vectors that were either chosen from a Normal distribution or by permuting the weights assigned to PC-1 in each session. The latter is the more conservative measure. Projections of the neural responses onto these random coding directions render $S_{rand}(t)$. Specifically, the degree of leverage is effectively zero or very much reduced. These analyses are summarized in a new Supplementary Figure S10. The distributions of our test statistics (e.g., leverage on choice and RT) under the variants of the null hypothesis also support traditional metrics of statistical significance. Figure S10 (bottom row) also provides an approximate answer to the question: What degree of leverage and mediation would be expected for a theoretical decision variable? Briefly, we simulated 60,000 trials using the race model that best fits the behavioral data of monkey M. For any noise-free representation of a Markovian integration process, the leverage of an early sample of the DV on behavior would be mediated completely by later activity as the latter sample—up to the time of commitment—subsumes all variability captured by the earlier sample. We, therefore, generated $S_{sim}(t)$ by first subsampling the simulated data to match the trial numbers of each session. To evaluate a DV approximated from the activity of N Tconin neurons per session rather than the true DV represented by the entire population, we generated N noisy instantiations of the signal for each of the subsampled, simulated trials. The noisy decision variable, $S_{sim}(t)$ is the mean activity of these N noise-corrupted signals. The simulation is consistent with the leverage and incomplete mediation observed for the populations of Tcon neurons. For in additional details, see Methods, §Leverage of single-trial activity on behavior) and Supplementary Figure S10, caption. See also our response to item 1 of the Public Response.

*R*2.5 The analysis is performed across several signed coherence levels, with data detrended for each signed coherence and choice to enable comparison of fluctuations relative to the relevant baseline; are results similar for the different coherences?*

The results are qualitatively similar for individual coherences. There is less power, of course, because there are fewer trials. The analyses cannot be performed for coherences $\geq 12.8\%$ because there are not enough trials that satisfy the inclusion criteria (presence of left and right choice trials with $RT \leq 670$ ms). Nonetheless, leverage on choice and RT is statistically significant for 27 of the 30 combinations of motion strengths $< 12.8\% \times$ three signals (S_{ramp}, S_{PC1} and S_{Tin}) $\times$ behavioral measures (RT and choice) (RT: all $p < 0.008$, Fisher-z; choice: all $p < 0.05$, t-test). The three exceptions are trials with 6.4% coherence rightward motion, which do not correlate significantly with RT on leftward choice trials. Reviewer Figure 4 shows the results of the leverage and mediation analyses, using only the 0% coherence trials.

*R*2.6 (Figure 6) Additional analysis to strengthen the claim that Min represents the integrand and not the integral:*

a. Repeating the analysis in Figure 6d with the integral (cumulative sum) of the single-trial Min signals and instead observing a significant increase in leverage over time would be strong evidence for this interpretation. If you again see no increase, then it suggests that the activity of these units (while direction selective) may not be strongly yoked to behavior. This scenario (no increasing leverage of the integral of Min on behavior through time) also raises an intriguing alternative possibility: that the noise driving the 'diffusion' of drift-diffusion here may originate in the integrating circuit, rather than just reflecting the complete integration of noise in the stream of evidence itself.

b. Repeating the analysis in Figure 6d with the projection of the M subspace onto its own first PC (e.g. take the union of units {M_{right}, M_{left}} [our $\{M_{in}^{right}, M_{in}^{left}\}$], do PCA just on those units' single

trial activities, identify the first PC, and project those activities on that dimension to obtain SPC1-M.

c. Ameliorating the sample-size limitation by relaxing the criteria for inclusion in Min - performing the same analyses shown, but including all units with visual RFs overlapping the motion stimulus, irrespective of their direction selectivity.

a. Reviewer Figure 2a provides support for leverage of the integral on choice, and this leverage, like $S_{Tin}^{con}(t)$, increases as a function of time. The effect is present in all seven sessions that have both M_{in}^{left} and M_{in}^{right} neurons (all $p < 1e - 10$). However, as shown in panel b, the same integral fails

to demonstrate more than a hint of leverage on RT (all correlations are negative) and the magnitude does not vary as a function of time. We suspect—but cannot prove—that this failure arises because of limited power and the expected weak effect. Recall that the mediation analysis of RT is restricted to longer trials and that the correlation between the Min difference and the signal is less than 0.1 over the heatmap in Fig. 6e, implying that the Min difference explains less than 1% of the variance of S_{Tin}(*t*). We considered including Reviewer Figure 2 in the paper, but we feel it would be disingenuous (cherrypicking) to report only the positive outcome of the leverage on choice. If the editors feel strongly about it,

we would be open to including it, but leaving these analyses out of the revised manuscript seems more consistent with our effort to deemphasize this finding. In the future, we plan to record simultaneously from populations MT and LIP neurons (Min and Tin, of course) and optimize Min neuron yield by placing the RDM stimulus in the periphery. We also provide this information in response to Recommendation (6) above.

b. We tried the R's suggestion to apply PCA to the union of Min neurons M_{in}^{right} , M_{in}^{left} , fully

expecting PC1 to comprise weights of opposite sign for the right and left preferring neurons, but that is not what we observed. Instead, the direction selectivity is distributed over at least two PCs. We think this is a reflection of the prominence of other signals, such as the strong visual response and normalization signals (see Shushruth et al., 2018). In the spirit of the R's suggestion, we also established an "evidence coding direction" using a regression strategy similar to the Ramp CD applied to the union of Min neurons. The strategy produced a coding direction with opposite signed weights dominating the right and left subsets. The projection of the neural data on this evidence CD yields a signal similar to the difference variable used in Fig. 6e (i.e., signals that are approximately constant firing rates vs time and scale as a function of signed coherence). These unintegrated signals exhibit weak leverage on choice and RT, consistent with Figure 6d. However, the integrated signal has leverage on choice but not RT, similar to the integral of the difference signal in Reviewer Figure 2.

c. We do not understand the motivation for this analysis. We could apply PCA or dPCA (or the regression approach, described above) to the population of units with RFs that overlap the motion stimulus, but it is hard to see how this would test the hypothesis that direction-selective neurons similar to those in area MT supply the momentary evidence. As mentioned, we have very few Min neurons (as few as two in session 3). Future experiments that place the motion stimulus in the periphery would likely increase the yield of Min neurons and would be better suited to study this question. As such, we do not see the integrand-like responses of Min neurons as a major claim of the paper. Instead, we view it as an intriguing observation that deserves follow-up in future experiments, including simultaneous recordings from populations of MT and LIP neurons (Min and Tin, of course). We have softened the language considerably to make it clear that future work will be needed to make strong claims about the nature of Min neurons.

*R*2.7 Other questions: Figure 2c is described as showing the average firing rate of units in Tconin on single trials, but must also incorporate some baseline subtraction (as the shown traces dip into negative firing rates). What baseline is subtracted? Are these residual signals, as described for later figures, or is a different method used? (Presumably, a similar procedure is used also for Figure 2a/b, given that all single-trial traces begin at 0.). Is the baseline subtraction justified? If the dataset really does reflect the decision variable with single-trial resolution, eliminating the baseline subtraction when visualizing single-trial activity might actually help to make the point clearer: trials which (for any reason) begin with a higher projection on the particular direction that furnishes the DV would be predicted to reach the decision bound, at any fixed coherence, more quickly than trials with a smaller projection onto this direction.*

We thank the reviewer for this comment. For each trial, the mean activity between 175 ms and

225 ms after motion onset was subtracted when generating the single-trial traces. The baseline subtraction was only applied for visualization to better portray the diffusion component in the signal. Unless otherwise indicated, all analyses are computed on non-baseline corrected data. We now describe in the caption of Figure 2 that "For visualization, single-trial traces were baseline corrected by subtracting the activity in a 50 ms window

around 200 ms.” Examples of the raw traces used for all follow-up analyses are displayed in Reviewer Figure 6.

Reviewer 3 ✖, private recommendations

I only have a few comments to make the paper more accessible:

*R*3.1 I struggle to understand how the linear fitting from -1 to 1 was done. More detail about how the single cell single-trial activity was generated to possibly go from -1 to 1 or do I completely misunderstand the approach? I assume the data standardization does that job?*

We have rephrased and added clarifying detail to the section describing the derivation of the ramp signal in the Methods (Ramp direction).

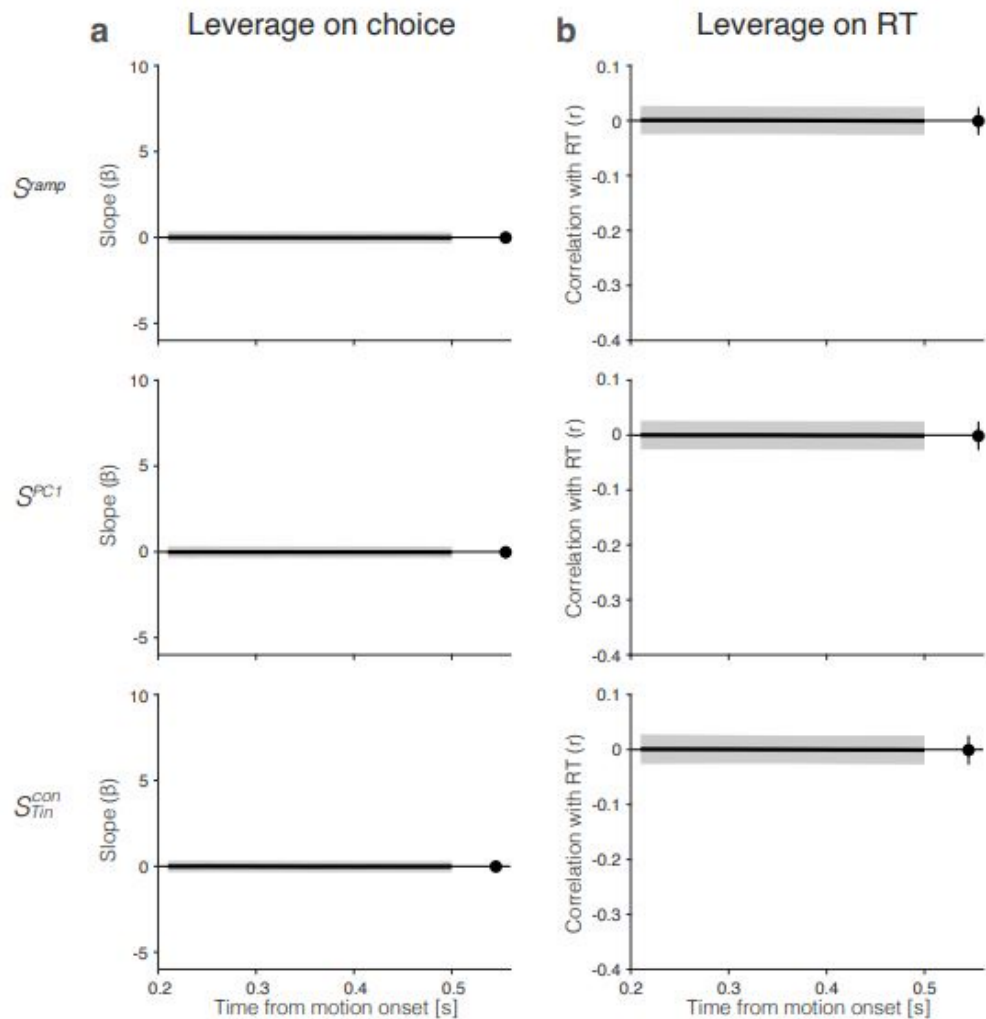
We applied linear regression to generate a signal that best approximates a linear ramp, on each trial, i , that terminates with a saccade to the choice-target contralateral to the hemisphere of the LIP recordings. The ramps are defined in the epoch spanning the decision time: each ramp begins at $f_i(t_0) = -1$, where $t_0 = 0.2$ s after motion onset, and ends at $f_i(t_1) = 1$, where $t_1 = t_{\text{sac}} - 0.05$ s (i.e., 50 ms before saccade initiation). The ramps are sampled every 25 ms and concatenated using all eligible trials to construct a long saw-tooth function (see Supplementary Figure S2). The regression solves for the weights assigned to each neuron such that the weighted sum of the activity of all neurons best approximates the saw-tooth. We constructed a time series of standardized neural activity, sampled identically to the saw-tooth. The spike times from each neuron are represented as delta functions (rasters) and convolved with a non-causal 25 ms boxcar filter. The mean and standard deviation of all sampled values of activity were used to standardize the activity for each neuron (i.e., Z-transform). The coefficients derived by the regression establish the vector of weights that define S_{ramp} . The algorithm ensures that the population signal $S_{\text{ramp}}(t)$, but not necessarily individual neurons, have amplitudes ranging from approximately -1 to 1.

*R*3.2 It is difficult to understand how the urgency signal is derived, to then generate fig S4.*

The urgency signal is estimated by averaging $S_x(t)$ at each time point relative to motion onset, using only the 0% coherence trials. We have clarified this in the caption of Supplementary Figure S4.

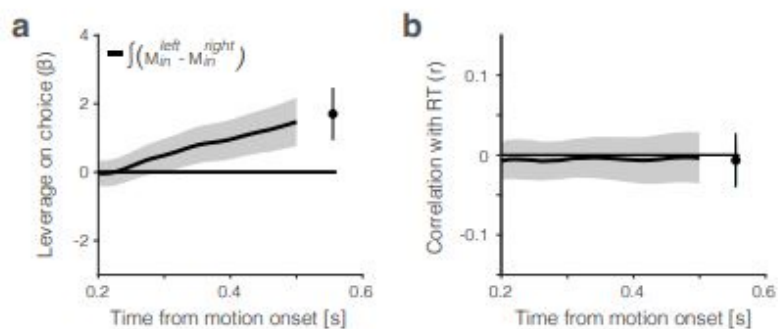
Author response image 1.

Shuffle control for Fig. 5. Breaking the within-trial correspondence between neural signal, $S(t)$, and choice suppresses leverage to near zero.



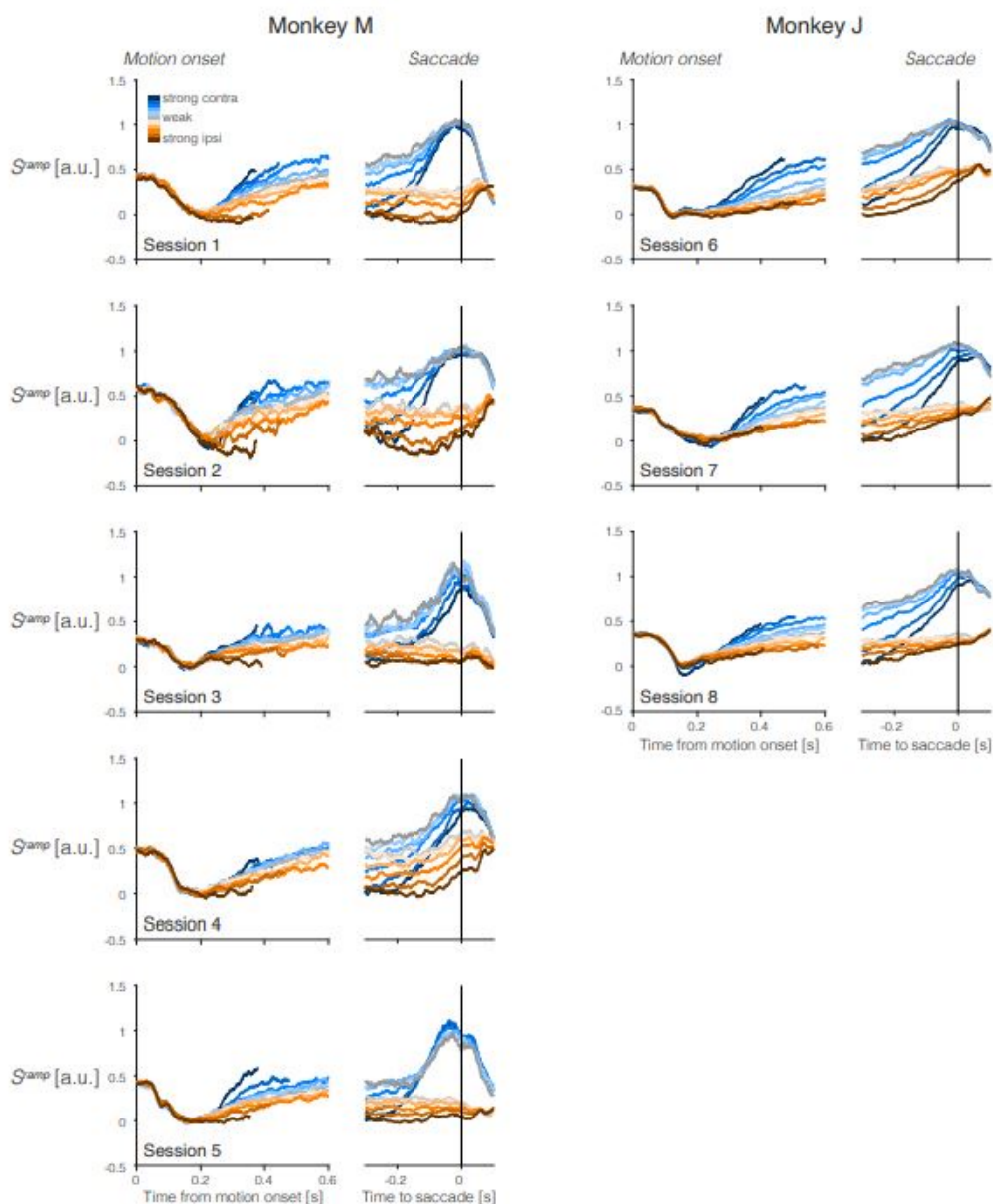
Author response image 2.

Leverage of the integrated difference signal $M_{in}^{left} - M_{in}^{right}$ on choice and RT. Traces are the average leverage across seven sessions. Same conventions as in Figure 5.



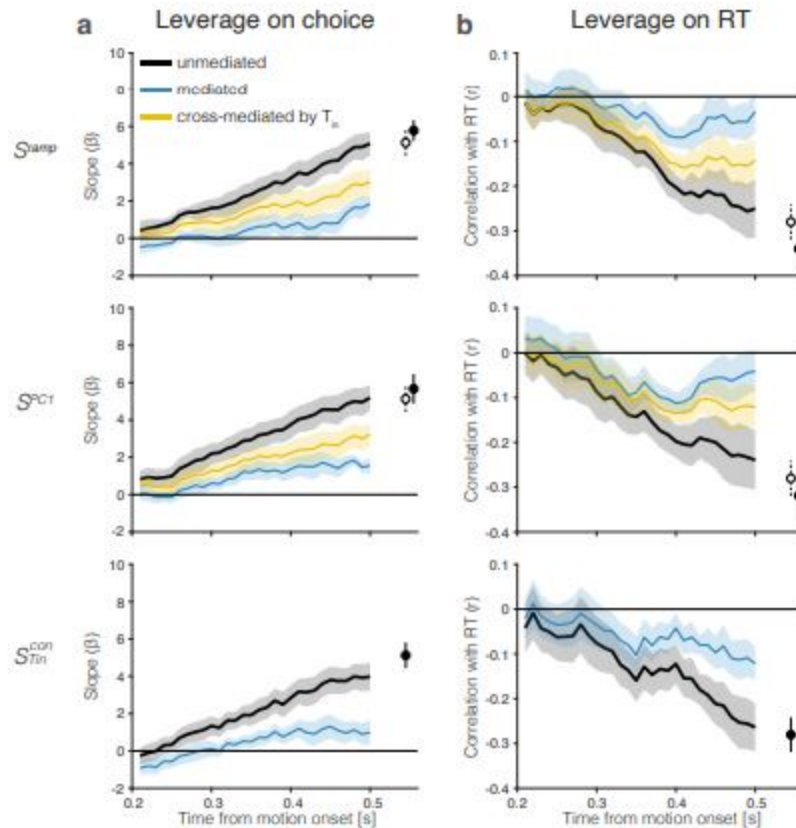
Author response image 3.

Trial-averaged S_{ramp} activity during individual sessions. Same as ?? for individual sessions for Monkey M (left) and Monkey J (right). The figure is intended to illustrate the consistency and heterogeneity of the averaged signals. For example, the saccade-aligned averages lose their association with motion strength before left (contra) choices in sessions 1, 2, 5, and 6 but retain the association in sessions 3, 4, 7, and 8.



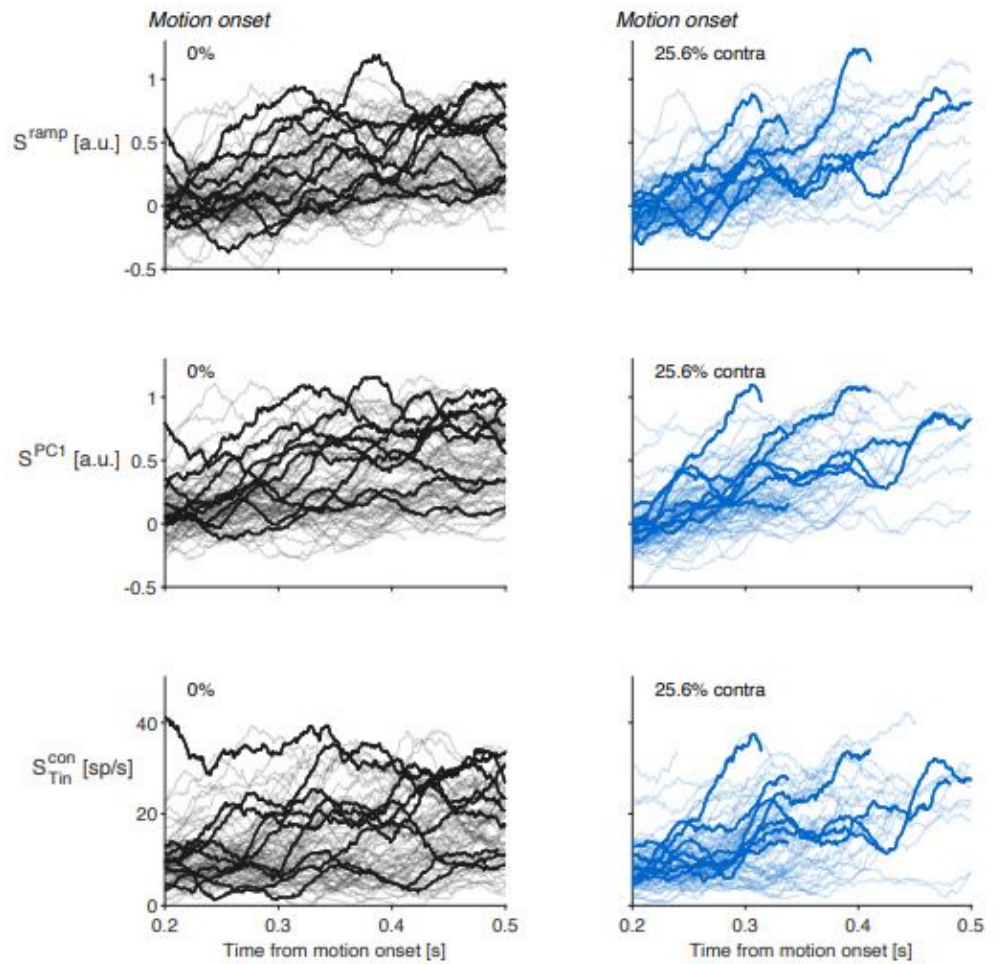
Author response image 4.

Drift-diffusion signals have measurable leverage on choice and RT even when only 0%-coherence trials are included in the analysis.



Author response image 5.

Raw single-trial activity for three types of population averages. Representative single-trial activity during the first 300 ms of evidence accumulation using two motion strengths: 0% and 25.6% coherence toward the left (contralateral) choice target. Unlike in Figure 2 in the paper, single-trial traces are not baseline corrected by subtracting the activity in a 50 ms window around 200 ms. We highlight a number of trials with thick traces and these are the same trials in each of the rows.



<https://doi.org/10.7554/eLife.90859.2.sa0>